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RESEARCH ARTICLE

# Enhancing the Early Student Dropout Prediction Model Through Clustering Analysis of Students' Digital Traces

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**ABSTRACT** Educational Data Mining and learning analytics have gained significant prominence in recent years, garnering attention from researchers worldwide. This is primarily due to their potential to improve decision-making processes within higher education. This study utilizes Educational Data Mining to suggest a clustering-based approach for classifying students based on their academic performance, specifically focusing on their interactions within Learning Management Systems in blended learning settings. Log data from the Learning Management System Moodle platform was analyzed to determine if valuable insights could be derived from student activities without the need for personal or invasive data. Given the large dataset, Principal Component Analysis and Uniform Manifold Approximation and Projection were employed for dimensionality reduction. Three unsupervised learning models—BIRCH, DBSCAN, and GMM—were then applied to identify distinct student clusters. The results from the comparison revealed that the BIRCH algorithm was the most effective in accurately categorizing students based on their activities. Further analysis of identified clusters revealed a strong correlation between student engagement and academic outcomes, with two high-risk clusters demonstrating the highest dropout rates. To better understand the reasons behind the found clusters, the course duration was subsequently divided into intervals, and standard machine learning methods were applied to predict at-risk students over time, confirming that early identification of at-risk students using clustering is feasible. This study demonstrates the efficacy of non-invasive data collection and unsupervised learning models in predicting student success, emphasizing the importance of temporal and content-specific analysis. The proposed approach leverages clustering algorithms to enhance the understanding of student behavior by identifying patterns and grouping students with similar learning and engagement profiles. This supports the development of targeted interventions that are tailored to specific student needs, ultimately improving retention and success.

**INDEX TERMS** Educational data mining, learning analytics, clustering analysis, dropout prediction.

## I. INTRODUCTION

Educational data mining (EDM) and learning analytics (LA) have been recognized as valuable research disciplines for examining and predicting the behavior and performance of current and future students. Furthermore, a significant rise in the volume of data, improved data formats, technology developments, and the accessibility of sophisticated analytical tools have all contributed to the growing use of LA and EDM methods [1]. Predicting the dropout rate and students' academic performance holds significant importance in higher education, particularly within universities.

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However, many existing studies have relied heavily on supervised learning techniques or the use of sensitive personal data, which may not be ideal for all educational contexts. There is a need for non-invasive methods that focus solely on behavior-related data and can still yield accurate predictions.

Higher education institutions are forced to continually increase the quality of education despite the unstable economic situation and societal changes, especially caused by the COVID-19 pandemic. Maintaining enrollment levels through program completion is vital for the sustainability of higher education institutions, which further emphasizes the need for data-driven approaches to understand and address student retention challenges. Consequently, there is an imperative need to employ innovative techniques that can provide deeper insights into student engagement and behavior and predict academic outcomes more accurately. This research addresses this gap by exploring how unsupervised machine learning techniques can be applied to data derived solely from students' interactions within Learning Management Systems (LMS), offering a non-invasive way to predict academic outcomes and identify at-risk students.

This research is motivated by the need to address the challenges faced by higher education institutions, including student dropout and the accurate prediction of academic performance. Machine learning techniques have increasingly been utilized to predict student performance at various educational levels. Traditional methods have often relied on supervised machine learning techniques, which, while helpful, may not fully capture the complex dynamics of student behavior in online learning environments [2], [3], [4].

In contrast, this study focuses on unsupervised learning techniques, which have been less frequently applied in educational settings and offer the potential to uncover hidden patterns in student behavior without the need for predefined labels. The non-invasive approach to student data analysis is ensured by leveraging data derived solely from online learning systems. The primary emphasis of this article lies in examining the potential of student interactions inside these platforms to yield predictive insights while eliminating the need to analyze sensitive demographic or extensive personal data. More precisely, this article describes a study that examines log files stored in the LMS Moodle to determine if valuable insights could be gained from student activities without using personal or invasive data. Given the large dataset size and the nature of log files, the data was preprocessed using standard techniques. Subsequently, Principal Component Analysis (PCA) was used for dimensionality reduction, followed by the non-standard approach of Uniform Manifold Approximation and Projection (UMAP), which was applied to reduce further the data dimensions. This approach was chosen to capture intricate structures, such as patterns of student engagement, performance trends, or behavioral similarities, which may not be easily detected using traditional methods, thereby enhancing the clustering process and providing deeper insights into student behavior.

The use of three distinct clustering algorithms—DBSCAN, GMM, and BIRCH—offers a unique contribution to the domain, as these methods have not been extensively compared in the context of EDM. By evaluating these models on LMS log data, this study demonstrates how clustering techniques can reveal hidden student behavior patterns and provide actionable insights for educational interventions. Each algorithm was applied to the dataset in the modeling phase, with the number of clusters determined using the elbow method and silhouette score to compare these algorithms objectively. Rather than relying on a single method, the comparison of these three algorithms offers a more robust analysis of the advantages and trade-offs of each clustering technique, ensuring that the findings are both comprehensive and generalizable.

To predict at-risk students as early as possible, the course duration was divided into four intervals: Q1 (first three weeks of the semester), Q2 (6 weeks), Q3 (9 weeks), and Q4 (13 weeks). Standard supervised machine learning methods were applied, including Decision Trees (DT), Support Vector Machines (SVM), Logistic Regression (LR), and Random Forests (RF) with tuned hyperparameters, to determine if it would be possible to identify at-risk students timely.

This research distinguishes itself from previous studies by employing unsupervised learning algorithms and working exclusively with LMS log data, thus avoiding the need for personal or sensitive information and invasive data collection methods. It does not aim to gather personal information but instead assesses the analytical significance of individuals' behaviors in virtual educational settings to predict academic results. Additionally, this study incorporates a temporal dimension, enhancing the understanding of how student behavior evolves over time and impacts academic outcomes. By focusing on the temporal dimension of student behavior, this study adds another layer of insight into how patterns of engagement evolve throughout the semester, providing early warning signals for at-risk students.

This study utilizes datasets derived from online learning systems to showcase the efficacy of clustering algorithms in identifying patterns that can be employed to predict student success. The findings underscore the importance of time-sensitive analysis and the contextual understanding of student interactions, offering a more nuanced perspective on how engagement metrics relate to academic results.

The contributions of this article can be summarized as follows:

- Educational data was analyzed using three sophisticated unsupervised machine learning models, which exhibit notable distinctions from the prevailing supervised and unsupervised machine learning approaches.
- A comprehensive temporal and content analysis of identified clusters of student engagement was realized in a virtual learning environment. The correlation with academic performance was shown, providing a foundation for targeted educational interventions.

- A prediction model using supervised machine learning techniques to identify at-risk students in various temporal intervals of the semester was developed and evaluated. Based on the results of previous students' data clustering, the feasibility of early intervention was demonstrated.

This paper's remaining sections are organized as follows. The second section reviews previous studies focusing on applying clustering techniques to predict student performance in higher education. It highlights significant findings and identifies gaps that this research aims to address. Section III applies the CRISP-DM framework, detailing each of its six phases. This section defines the research objectives, involves data description, and covers all activities needed to create the final dataset from the LMS Moodle. It also describes the clustering model tunning and development using the BIRCH, DBSCAN, and GMM algorithms. Those models are assessed to ensure they meet the requirements set in research objectives. Section IV presents the outcomes of the clustering analysis, detailing how student groups were identified based on their interactions within the learning platforms. Section V develops and evaluates a classification model based on the clusters identified in the previous section to predict student performance. Section 6 analyzes the implications of the findings, the effectiveness of the clustering and classification models, and their potential impact within real-world education settings. In contrast, Section 7 summarizes the study's conclusions, discusses its contribution to the field of EDM, and outlines possible directions for future research.

## II. RELATED WORK

The related work in EDM and LA showcases a broad spectrum of methodologies and applications, including supervised and unsupervised machine learning methods, aimed at enhancing the teaching-learning process through data-driven insights.

A notable study [5] utilized data mining techniques to analyze educational data from a higher education institution in Croatia, focusing on e-course event logs. This study applied cluster analysis to categorize student behavior based on similarities in their interactions with course materials. Moreover, it utilized decision trees to create decision-making models that deepen the understanding of student learning behaviors. This approach emphasized the potential of pattern recognition and predictive modeling in identifying and addressing educational needs.

Further, a comprehensive review [6] spanning over a decade (2010-2022) discussed the evolution and application of EDM techniques across 142 research articles sourced from prominent databases like IEEE and Elsevier. The review highlighted a predominant use of classification techniques, often combined with clustering, association rule algorithms, and regression techniques. Such combinations have proven effective in classifying and analyzing student performance, predicting dropout rates, and enhancing instructor performance. This extensive analysis underscored the ongoing advancements in EDM and its critical role in optimizing educational strategies based on various data-driven techniques.

Another study [7] explored the integration of convolutional neural networks (CNNs) with k-means clustering and discriminant analysis to predict students' academic performance. This innovative approach introduced a novel statistic to optimize the determination number in k-means, enhancing the clustering reliability and, subsequently, the accuracy of performance predictions.

Bardach et al. [8] examined how context-related problems and motivational factors like personal best goals influence university students' dropout intentions. This study underscored the importance of considering structural and individual factors in dropout prediction models.

Similarly, Xing and Du [9] demonstrated the potential of educational games as predictive tools for academic performance, showcasing the versatility of machine learning applications in education. Additionally, Mubarak et al. [10] emphasized the significance of temporal stability in learning behaviors when predicting student dropout, which aligns with the approach of analyzing student performance across specific timeframes in this research. Finally, Fei and Yeung [11] highlighted how procrastination and learning strategies impact student success in blended learning environments, further supporting the importance of behavioral analysis in predictive modeling.

Moreover, technology's role in data analysis within education has become crucial for assessing student achievement and behavior patterns. Researchers could use state-of-the-art models like decision trees and DBSCAN for clustering to predict educational institutes' performance with higher accuracy. These techniques helped in understanding not just the outcomes but also the influencing factors behind student behavior and achievement, highlighting the multifaceted benefits of EDM in educational settings [12].

Vital et al. [13] used k-means and hierarchical machine learning techniques for student performance cluster analysis cluster analysis. They demonstrated that the methods used are reliable for predicting student academic success. They showed essential contributing elements through linkages in dendrograms and hierarchical clustering that affect student results, such as extracurricular activity, attendance, and the number of failed classes.

Similarly, Govindasamy and Velmurugan [14] compared four clustering algorithms: fuzzy c-min, k-means, k-medoids, and expectation maximization (EM). The study used data from 131 college students to predict the student's performance on the semester's final exam. According to the study's findings, fuzzy c-means and EM required more time to install but produced superior clustering quality in terms of purity and normalized mutual information.

Another study [15] focused on the issue of delayed thesis completion in undergraduate programs in a more targeted analysis. Employing various clustering algorithms, including

k-means and k-medoids, this research correlated students' Grade Point Averages and course grades with thesis completion times, uncovering moderate correlations that suggest higher academic performance correlates with faster completion. This insight not only aids in understanding factors that delay graduation but also supports institutions in guiding students more effectively through their studies.

Further exploring the integration of digital technologies in education, a study [16] on student engagement in a Blended Learning (BL) environment applied the SimpleKmeans clustering technique to identify diverse student engagement patterns. This approach demonstrated the variability in how students interact with learning resources, emphasizing the need for personalized educational strategies to optimize learning outcomes.

A study [17] analyzing the academic performance of business students in Malaysia focused on the shift to online learning during the COVID-19 pandemic. Utilizing k-means clustering, the research revealed an improvement in students' Cumulative Grade Point Average during the pandemic, suggesting that the new learning modalities might have positively impacted their academic achievements.

Lastly, a study [18] that benchmarked algorithms like k-means, hierarchical clustering, and fuzzy C-means against multiple internal validity metrics analyzed the effectiveness of various clustering algorithms. This evaluation was pivotal in determining the most suitable algorithms for mining knowledge from student engagement data, showcasing the need to tailor data mining approaches to specific educational contexts for optimal outcomes.

However, dropout rates are not caused by a single factor [8]. Instead, they depend on a variety of factors and differ throughout universities. This study will look at students' behavioral characteristics in clustering their performance in addition to factors obtainable from learning management systems.

## III. METHODOLOGY AND RESULTS

The CRISP-DM (Cross Industry Standard Process for Data Mining) methodology was selected as pivotal to this study because it guided the development of clustering techniques alongside predictive models as a standard approach in the knowledge discovery process. This methodology is for academic research, including EDM and LA domains, due to its structured approach to tackling complex data-driven problems [19]. Figure 1 shows the conceptual framework of the research.

The analysis was conducted using Python 3.8 within the Anaconda distribution, a popular choice for data science due to its comprehensive suite of data-centric Python packages. The packages were utilized as follows:

- *Data Management: Pandas* were used to import and manage data frames, facilitating efficient data manipulation and cleaning.
- *Array Computing: NumPy* was employed for high-performance array computing, which is crucial for handling large datasets typically involved in student performance analysis.
- *Machine Learning Models: Scikit-learn* provided the machine learning algorithms necessary for clustering and classification tasks, thanks to its robust, well-documented libraries.
- *Data Visualization: Matplotlib* was used to generate static plots, and *Seaborn* was used for more advanced statistical visualizations to aid in interpreting data and communicating findings.

This combination of Python libraries and tools enabled the robust analysis and insightful visualization of complex patterns in student engagement and performance, which is integral to developing actionable insights in educational settings.

### A. BUSINESS UNDERSTANDING

Analyzing student behavior and academic performance data is crucial to understanding why dropout rates are high, particularly in higher education institutions [8]. Similarly, to many other universities, failures in exams and a lack of interest in the courses are the main reasons for student dropouts at Constantine the Philosopher University in Nitra.

In this study, patterns in student engagement are identified through LMS logs to explore their correlation with academic performance, aiming to reduce dropout rates and improve graduation metrics. After identifying these patterns, clustering algorithms are used to group students with similar engagement behaviors. Subsequently, classification methods, including Logistic Regression, Random Forest, SVM, Decision Trees, and K-Nearest Neighbors, are applied to assign individual records to the identified clusters. This combined approach supports academic planning and decision-making by enabling targeted interventions.

Previous research [20], [21], [22] indicated that meaningful analytical results are contingent upon the full utilization of virtual learning environments. This presented study gains valuable insights into students' course persistence by analyzing anonymized course log data, which captures student behavior such as assignment submissions, quiz or test performance, and interaction with course materials. This approach ensures that the analysis respects student privacy while providing actionable insights by identifying patterns that could indicate potential risks of dropout or disengagement.

The semester was organized into four quarters over 13 weeks, providing several occasions to assess student engagement without making the course overly dependent on activities directly tied to their final grades. Students engaged in several optional tasks, quizzes, and activities and could participate in lecture-based open-ended questions throughout each quarter. The only semi-mandatory activity, a mid-term test in the ninth week, was not a requirement for advancing to the final examination. This flexible course structure was

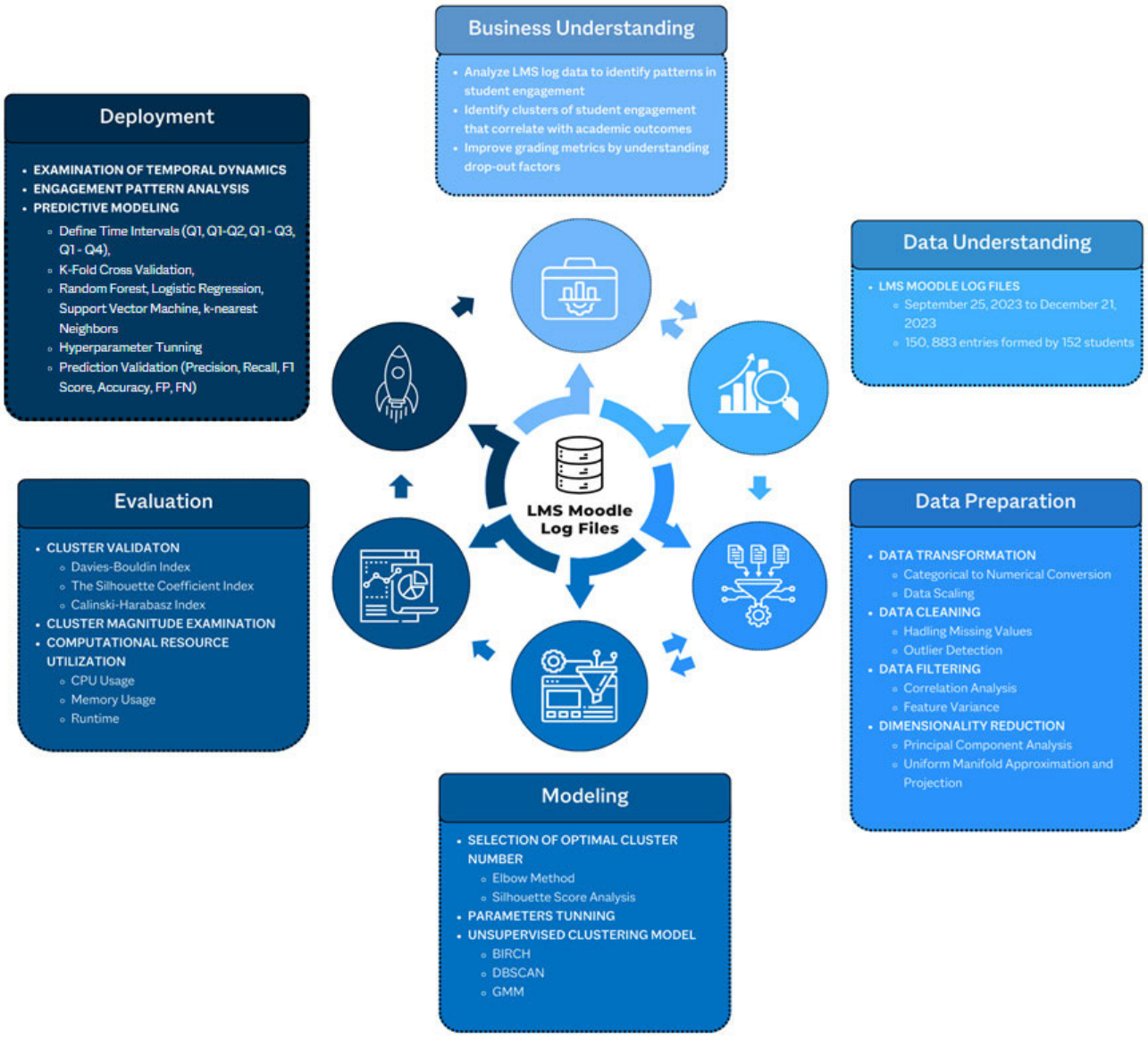



**FIGURE 1.** Conceptual methodological framework based on CRISP-DM methodology used in the study.

designed to encourage natural student engagement, providing a richer dataset for the selected unsupervised learning algorithms to analyze. The primary aim of this research was to explore the potential of unsupervised learning algorithms in interpreting engagement data obtained from LMS Moodle log files for the examined course, which record various student activities across the course, and their subsequent impact on academic performance. Furthermore, the aim was to identify whether any of the algorithms could successfully cluster students into meaningful groups based on their engagement data, with the goal of exploring how these clusters correlate with their academic outcomes. The intent was to provide educators with a data-driven approach to better understanding and enhancing student engagement in blended learning environments.

### *B. DATA UNDERSTANDING*

The dataset used in this research was obtained from the course opened at Constantine the Philosopher University in Nitra. The course consisted of online activities of 152 university students enrolled in a blended Computer Science course. This course combined traditional lectures and lab sessions with online interactions and online access to course materials during the winter semester from September 25, 2023, to December 21, 2023. LMS Moodle served as a central virtual learning platform for this course. Students could access digital course materials, engage in collaborative projects, submit assignments, participate in quizzes, and take active responsibility for their learning journey in line with the self-regulated learning paradigm.

**TABLE 1. Brief description of Moodle logs attributes.**

| Attribute | Description |
|---|---|
| *Timestamp* | The date and time when the activity record was created. |
| *User Full Name* | The name of the user engaged in the activity. |
| *Component* | The name of the course activity. |
| *Action* | A variable that describes a specific action performed within the component. |
| *Target* | The target variable specifies the object or entity that is the target of the given action. |
| *Description* | A brief description of the activity. |
| *IP Address* | The internet protocol address used to access the LMS. |

Moodle's built-in logging system was used to meticulously record each student's course interactions, including activities such as quiz attempts, assignment submissions, forum participation, and access to course materials. The generated log files included detailed data on each student's identity, the IP address used for each session, precise timestamps of activities, and descriptions of the actions taken. Over the 13-week academic semester, these records amounted to 150,883 entries, forming a comprehensive dataset for analysis. The log file structure, including several key elements, is described in Table 1.

This dataset, comprising anonymized log data from Moodle, offered a unique opportunity to analyze various aspects of student behavior and interaction with the course. The analysis aimed to uncover patterns in online engagement—such as quiz performance, assignment submissions, forum participation, and resource access—and assess the impact of these activities on student performance. By applying clustering algorithms, distinct student behavior groups were identified, providing valuable insights into the effectiveness of blended learning environments and helping to tailor interventions for improved academic outcomes.

### C. DATA PREPARATION

Effective data pre-processing is crucial to ensure the integrity and suitability of the dataset for subsequent clustering algorithms. The designed methodology encompasses five key pre-processing techniques:

*Data Transformation:* In this phase, converting all categorical variables to numeric values was crucial. After addressing the categorical variables, the next step was data normalization or scaling to ensure uniformity in measurement scales. These conversions were essential, especially within algorithms that compute distances between data points, such as clustering algorithms.

*Data Cleaning:* This stage involved removing or correcting erroneous data entries, handling missing values, and eliminating outliers. This process was essential to avoid skewed results and improve the reliability of the clustering outcomes.

*Attribute Selection:* It was focused on identifying and retaining the most significant attributes that contribute to the analysis while discarding redundant or irrelevant ones. It was based on their predictive ability regarding student engagement and performance.

*Dimensionality Reduction:* Although not previously mentioned, this technique is often vital in managing datasets with many variables. It simplifies the input dataset by reducing the number of dimensions without losing important information.

Each of these stages plays a vital role in refining the dataset, enhancing its structure and quality, and ensuring that it is optimally prepared for effective analysis using clustering algorithms.

#### 1) DATA TRANSFORMATION

Most machine learning algorithms generally require numeric input and output variables. This restriction must be addressed when implementing and developing machine learning models. This implies that all characteristics, including categories or nominal variables, must be transformed into numeric variables before entering data into the clustering and classification model [23].

Therefore, in the process of data transformation, the variable *Component* was mapped to the numerical values: *Assignment*: 0, *Attendance*: 1, *Course*: 2, *Project*: 3, *Quiz*: 4, *Study_material*: 5. Similarly, *Action* and *Target* were also converted into numerical representations using predefined mappings, thus enabling these categorical attributes to be processed by various machine learning algorithms.

Machine learning models are designed to identify relationships between input variables (features) and output variables (predictions or target outcomes). This means the model uses input data, such as student engagement metrics, to predict specific outcomes, such as academic performance or likelihood of course completion. However, the scale and distribution of the domain data may differ for each variable. Moreover, input variables may have distinct units, which means they may have different scales. Differences in scaling among input variables may increase the complexity of the modeled problem. Large input values can result in a model that learns large, weighted values. A model with weighted values is frequently unstable, so it may perform poorly during learning and be sensitive to input values, resulting in a larger error. On top of that, normalization was performed using the StandardScaler or z-transformation and MinMaxScaler methods to form datasets for three different models. The first method, StandardScaler, allowed each attribute's values to be in the same range so that a comparison could be made. Z-transformation normalization refers to normalizing every value in a dataset such that the mean of all values is 0 and the standard deviation is 1. In the second method, MinMaxScaler converted all attributes into a range [0, 1], which means the minimum value of the attribute was zero, and the maximum value of the attribute was one [24].

As mentioned before, the intention was to apply BIRCH, DBSCAN, and GMM in the modeling phase. The choice between StandardScaler (Z-transformation) and MinMaxScaler (normalization) depends on the data and the context of the problem. BIRCH and DBSCAN are sensitive to the scale of the data since they rely on the distance between data points. The GMM assumes that the data is Gaussian. Therefore, it benefits from scaling that keeps this assumption intact. Considering those facts, StandardScaler was used to maintain the data structure.

Subsequently, several additional time-related attributes were extracted from the timestamp attribute to provide deeper insights into the temporal patterns of student interactions. These included *weekday*, *week*, and *month*. Such detailed temporal information could help to understand student engagement trends over time.

Ensuring the privacy of student data was paramount. Hence, each student's data was anonymized. This step was critical to maintaining confidentiality and compliance with data protection regulations while still allowing individual tracking across the dataset.

The attributes *IP address* and *description* were removed from the dataset. The rationale for omitting the IP address was that it did not contribute to the academic analysis; its primary function of identifying users, locations, or devices was not relevant to the study's objectives, which focused on educational outcomes. Similarly, while providing narrative context, the description field did not offer quantifiable data for clustering or pattern recognition and was thus deemed non-essential for the analytical processes. These pre-processing steps ensured the dataset was clean, well-structured, and aligned with the study's objectives to explore the impact of various educational engagements on student performance.

### 2) DATA CLEANING

The second data preparation phase involved rigorous checks for missing values and outliers within the log files. Every entry in the dataset was checked, ensuring completeness and consistency across all data points. This examination confirmed the absence of missing entries, which was crucial for maintaining the integrity of subsequent analyses. Outlier analysis was another critical component of this stage. Statistical techniques were utilized to identify any extreme values within numerical attributes that could potentially skew the results of the selected clustering algorithms.

The results of this analysis are visually represented in Figure 2 using a box plot. The box plot depicts the distribution of data points, highlighting the minimum, maximum, quartiles, and any potential outliers. The analysis indicated that no extreme outliers were present, affirming the robustness of the dataset for detailed analysis. These initial steps were vital for ensuring that the dataset was complete and representative of the proper patterns and behaviors within the student population, thereby enhancing the reliability of the obtained findings from clustering algorithms.

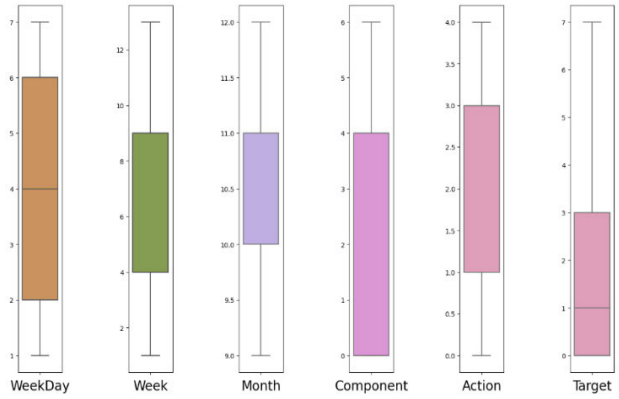



**FIGURE 2.** Box plot of numerical variables for outlier analysis.

### 3) DATA FILTERING (ATTRIBUTE SELECTION)

Too many attributes in educational datasets can cause dimensionality issues and difficulties when processing and analyzing data. Moreover, the calculation of distance by clustering algorithms may not be effective for high-dimensional data. To address this, feature selection methods were applied to identify the most relevant attributes for this study.

The feature selection process began with a correlation matrix to examine the pairwise relationships between numerical variables in the dataset. Values close to 1 or -1 indicate a strong positive or negative correlation, respectively, suggesting that as one variable increases, the other variable tends also to increase (positive) or decrease (negative). Values around 0 suggest no linear correlation.

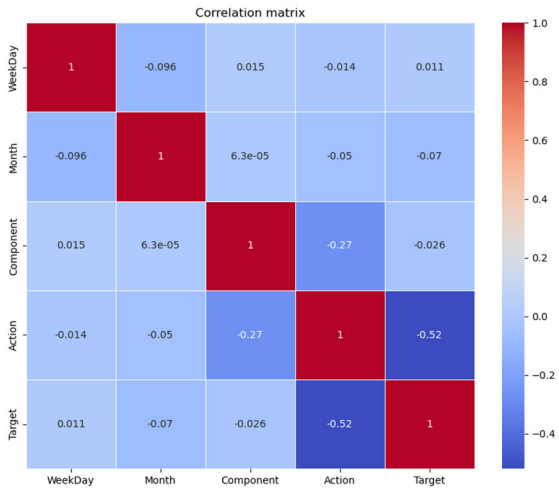



**FIGURE 3.** Correlation matrix of numerical variables.

The correlation matrix (Figure 3) suggested that most variables have a low correlation, implying little redundancy between them. This is beneficial when performing clustering as it reduces the risk of multicollinearity affecting the results.

In addition, a variance threshold for feature selection was applied to ensure the effectiveness of the unsupervised machine learning model. Variance, as a metric, quantifies the

dispersion of a dataset distribution. It is pivotal in unsupervised learning, particularly in clustering, where identifying distinct patterns is crucial. Selecting attributes with considerable variance is essential to prevent a bias in the clustering model towards certain attributes that may not be as discriminative. Attributes with high variance demonstrate a wide range of values (high cardinality), suggesting they can contribute unique information for clustering.

Next, a variance threshold of 0.3 was applied to filter out low-variance attributes, ensuring the remaining features had sufficient variability to contribute meaningfully to the clustering process. This threshold was chosen based on its suitability for distinguishing relevant behavioral patterns within the data. The variance of each student attribute is illustrated in Figure 4. As can be seen, all attributes exceeded the variance threshold of 0.3. As a result, this indicated that other attributes exhibited significant variability and were likely to contribute valuable insights into student behavior patterns.

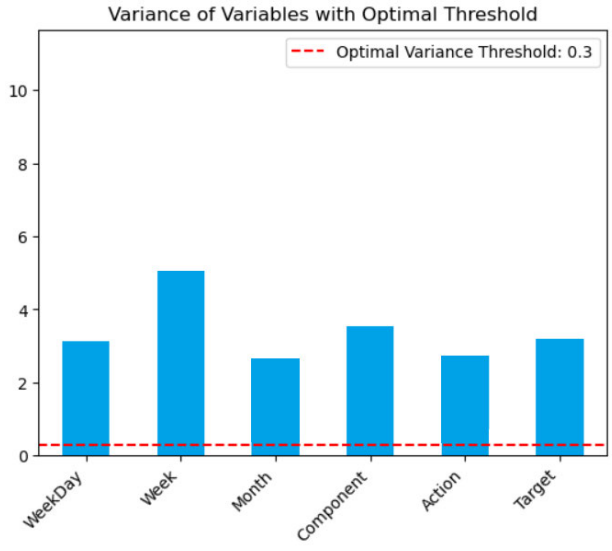



**FIGURE 4.** Variance threshold of numerical variables.

The steps taken for feature selection confirmed that the attributes retained were relevant, as the correlation results showed low redundancy between most variables, reducing the risk of multicollinearity and ensuring more reliable clustering. Additionally, the variance results indicated significant variability across all features, further supporting their potential to provide valuable insights into student behavior patterns.

#### 4) DiMENSIONALITY REDUCTION

In the data preparation phase of the analysis, dimensionality reduction was crucial for managing the computational complexity and enhancing the visualization of the examined educational dataset. Principal Component Analysis (PCA) was initially applied to reduce the dimensionality of the examined dataset while retaining 95% of the variance. This variance retention threshold dictated the number of components selected by PCA, effectively condensing the dataset into a lower-dimensional space that preserves the most informative features for clustering.

Figure 5 presents two graphs to illustrate the process of determining the number of principal components. Figure 5a, ''% Explained Variance,'' displays the individual contribution of each principal component to the dataset variance. It is evident that the four components account for the highest variance, with each subsequent component contributing progressively less.

On the other hand, Figure 5b, ''% Cumulative Variance,'' has a marked point labeled ''$CV = 0.954$'' at the fifth component, which implies that approximately 95.4% of the total variance has been captured by the first five principal components. This suggests that selecting six components would retain most of the information contained in the original dataset while substantially reducing its dimensionality.

Upon identifying the six optimal components through PCA, which accounted for 95.4% of the variance, Uniform Manifold Approximation and Projection (UMAP) was applied to visualize the data in a two-dimensional space. UMAP is renowned for its effectiveness in preserving both local and global data structures and provides an insightful representation for further analysis. The utilization of UMAP after PCA serves two main purposes. First, it simplifies the complexity of multi-dimensional data, making it more manageable for computational processes. Second, and most importantly for this analysis, it facilitates a more intuitive visualization of the inherent clustering of the dataset.

The UMAP model was configured with two components to reduce the high-dimensional data into a two-dimensional representation, making it easier to visually explore and interpret the complex relationships within the dataset. This dimensionality reduction helps uncover patterns and groupings that are not easily observable in the original multi-dimensional space, allowing for a clearer understanding of the underlying structure in the data. The parameter *n_neighbors* was set to 150, influencing UMAP's balance between local and global structure in the data. In contrast, *min_dist*, set to 0.5, controlled how tightly UMAP packs the points in the low-dimensional space. These specific settings were chosen to strike a balance between the granularity of the clusters and the broad arrangement of the data, allowing for clear differentiation between clusters while maintaining the overall data distribution. The resulting two-dimensional embedding crafted by UMAP captures the intrinsic geometry of the high-dimensional space and presents it in a way that is intuitive for human observers.

The UMAP embedding displayed in Figure 6 illustrates distinct clusters and densities, indicating variability in student engagement and activity levels. These groupings provide critical insights into how students interact within the learning environment, reflecting different learning styles or engagement strategies. The visualization helps pinpoint areas of high density and notable outliers, facilitating a deeper analysis in the clustering phase to tailor educational interventions effectively.

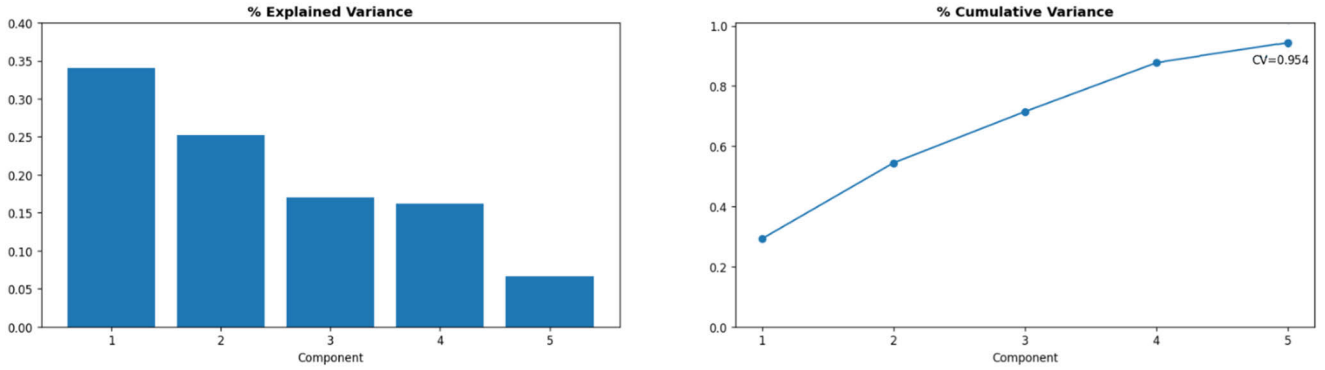



**FIGURE 5.** a) Explained variance by component b) Cumulative variance by component.

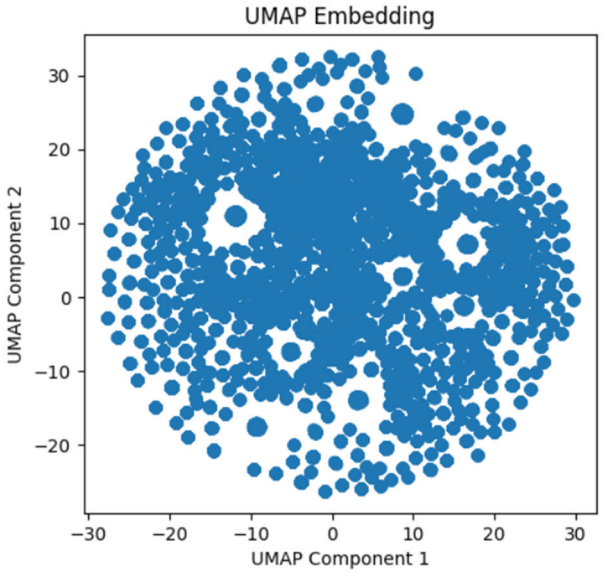



**FIGURE 6.** UMAP visualization of the dataset.

### D. MODELLING

In recent years, researchers have garnered significant attention to the use of clustering techniques for predicting student performance. Clustering involves grouping similar objects into one cluster while distinctly different objects are grouped into separate clusters. This technique proves particularly useful when labeled information within a student dataset is unknown. Moreover, dividing large datasets into smaller, logical clusters facilitates more straightforward examination and interpretation by researchers.

Among the various clustering methods, the k-means algorithm is often a preferred choice due to its simplicity in implementation and straightforward results. K-means groups the objects based on their proximity to the centroid of the clusters. The elbow method is commonly used to determine the optimal number of clusters. This technique involves calculating the total within-cluster variance—or inertia—for different numbers of clusters ($k$) and identifying the point where the variance curve initially levels off, indicating the most appropriate value for $k$.

However, the k-means algorithm, despite its popularity, has certain limitations, particularly for the type of educational data analyzed in this study. K-means assumes that clusters are spherical and equally sized, which may not always reflect the true structure of complex educational data. Additionally, it is sensitive to outliers and noise, which can skew the results when dealing with datasets like student interaction logs. More advanced clustering techniques, better suited for handling the nuances of the examined data, were explored due to these limitations.

The choice of BIRCH, DBSCAN, and Gaussian Mixture Models (GMM) was guided by several key considerations:

- BIRCH (Balanced Iterative Reducing and Clustering using Hierarchies): Is particularly effective for handling large datasets, such as the vast amount of interaction data generated in online learning environments. Its incremental and hierarchical nature allows it to process data in a memory-efficient manner, making it suitable for educational data that grows over time. Furthermore, BIRCH can automatically adjust to the available computational resources, ensuring that it delivers optimal clustering quality based on the dataset size. This is a critical advantage in educational settings where data is often multi-dimensional and complex.
- DBSCAN (Density-Based Spatial Clustering of Applications with Noise): One of DBSCAN's main advantages is its ability to discover clusters of arbitrary shape and handle noise effectively. This is especially useful when working with student engagement data, which often includes sparse or irregularly distributed patterns of interaction. DBSCAN's sensitivity to density allows it to differentiate between dense areas of student activity (indicating high engagement) and outliers, making it a powerful tool for identifying distinct groups of students while filtering out noise.
- GMM (Gaussian Mixture Models): Unlike algorithms such as k-means, which assume hard assignments (i.e., each data point belongs to one cluster), GMM provides

soft cluster assignments, meaning each data point is assigned to a cluster with a probability. This probabilistic approach is valuable for educational data where student behavior might not fit neatly into rigid categories. GMM allows to capture more nuanced relationships and latent variables within the data, offering a better understanding of student performance and engagement levels.

Other potential algorithms, such as hierarchical clustering and spectral clustering, were considered but ultimately excluded for the following reasons:

- Hierarchical clustering, while useful for smaller datasets, becomes computationally expensive and inefficient with larger datasets, such as the LMS log data used in this study. This method also struggles with handling noise, which is often present in real-world educational data.
- Spectral clustering was also considered, as it can handle non-linearly separable data. However, its high computational cost and sensitivity to scaling issues made it less suitable for the size and nature of the examined dataset.

In summary, the selection of BIRCH, DBSCAN, and GMM for this study was based on their ability to handle the high-dimensional, noisy, and complex nature of student engagement data. These algorithms offer distinct advantages that are well-suited to the unique challenges posed by educational data, making them a superior choice over more commonly used algorithms like k-means. By leveraging these advanced techniques, the aim is to uncover subtle patterns and relationships within student data that can inform targeted interventions. These methods enable the segmentation of student populations based on engagement, potentially leading to more effective educational strategies and improved academic outcomes.

### 1) SELECTING THE OPTIMAL NUMBER OF CLUSTERS

Determining the optimal number of clusters for clustering algorithms is a critical step in ensuring the meaningful segmentation of data. Two widely recognized methods, the elbow method, and the silhouette score analysis, were employed to ascertain the most appropriate clustering solution for the examined educational dataset. The Elbow method is a heuristic method for determining the number of clusters in a dataset. It involves plotting the explained variance as a function of the number of clusters and picking the elbow of the curve as the number of clusters to use. This method looks for the point at which the rate of decrease sharply changes, indicating that additional clusters do not significantly contribute to the explanation of variance.

The Elbow Method (Figure 7) illustrates that as the number of clusters increases, the within-cluster sum of squares (WCSS) decreases because the data points are closer to the centroids of their clusters. The point where the decline in WCSS starts to level off is aimed at indicating that adding more clusters does not significantly improve the variance explained by the model in the data.

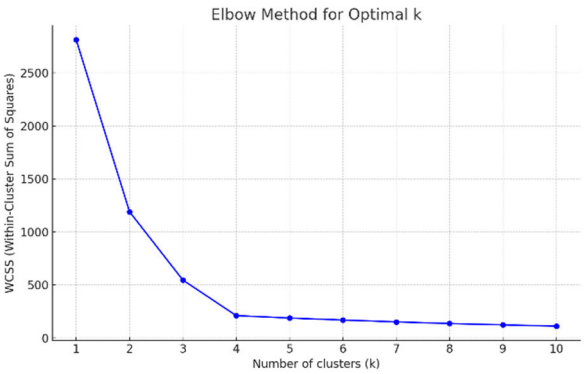



**FIGURE 7.** Elbow method for optimal k value.

The results indicated that the sharpest decrease in WCSS occurs between 1 and 4 clusters, with the curve flattening out beyond this point. This visual cue suggests that an appropriate number of clusters for these data could be around 4, as further increases in the number of clusters do not result in a substantial reduction in WCSS.

Following the elbow method, silhouette score analysis was utilized to validate the cluster count further. The silhouette score is a measure of how similar an object is to its own cluster compared to other clusters. The higher the silhouette score, the more appropriately the objects have been classified.

The silhouette analysis provided a clear insight into the cohesion and separation of the clusters formed in the proposed educational dataset. As depicted in the silhouette plots (Figure 8) for $k = 4, 5, 6, 7$, the width of the silhouette reflects how well each data point lies within its cluster. The aim was to find a balance between the highest average silhouette score and the most uniform silhouette width across all clusters. The silhouette score was approximately 0.3659 for $k = 4$, indicating a reasonably good structure given the complex nature of educational data. However, when increasing the number of clusters to $k = 5$ and $k = 6$, the silhouette score decreased slightly to 0.3483 and 0.3563, respectively, suggesting that the additional clusters might not capture more meaningful distinctions between student behaviors.

However, the silhouette score for $k = 7$ was 0.3691, marginally higher than for $k = 4$. However, when considering the visual consistency of the silhouettes across the clusters, $k = 4$ demonstrated a more balanced distribution with relatively uniform silhouette widths, which is preferable for the interpretability and reliability of the clustering results.

The optimal number of clusters for this research was established by a comprehensive approach, integrating the Elbow method with silhouette score analysis. The Elbow method suggested a natural clustering solution around 4 clusters, as indicated by the point at which the WCSS began to decrease at a slower rate. Further substantiation came from the silhouette analysis. Not only did a four-cluster solution yield a satisfactory silhouette score, but it also demonstrated a uniform distribution of silhouette widths. This uniformity

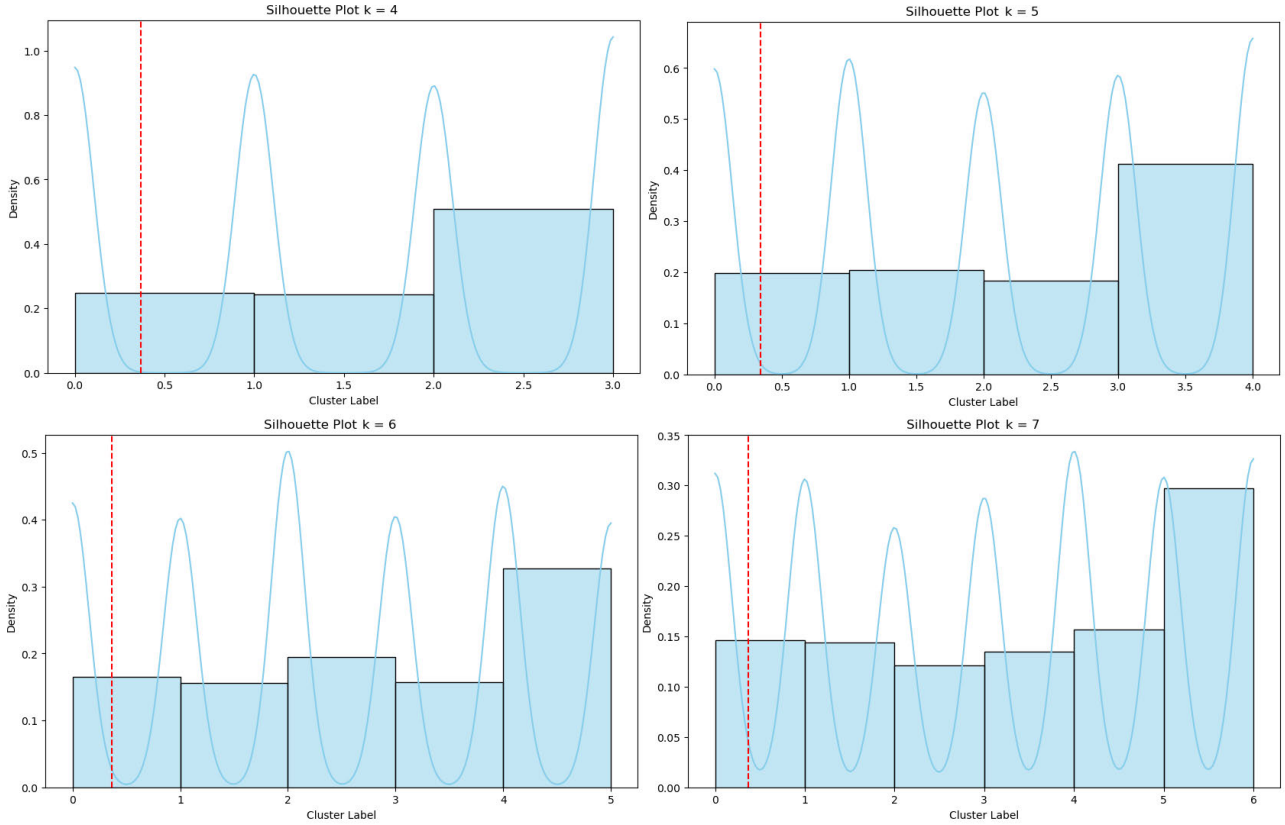



**FIGURE 8.** Silhouette score analysis plot for k = 4,5,6,7.

**TABLE 2.** Results of parameter tunning.

| Clustering Model | Parameter | Value | Best Parameters |
|---|---|---|---|
| *BIRCH* | *threshold* | *0.1 – 0.5* | 0.3 |
| | *branching factor* | *20 - 100* | 70 |
| *DBSCAN* | *Epsilon* | *0.1 – 1.0* | 0.4 |
| | *Min_samples* | *5 - 50* | 15 |
| *GMM* | *Covariance type* | *full, tied, diag* | full |
| | *max_ites* | *100 - 500* | 250 |

implies that all clusters are well-defined and significant to each other, with each cluster exhibiting strong internal cohesion and clear separation from other clusters.

### 2) HYPERARAMETERS TUNNING

Hyperparameter tuning is a critical process of machine learning. It involves selecting the optimal parameters for a learning algorithm to maximize its performance. Hyperparameters, which are values set before the learning process begins, significantly influence how the algorithm behaves and learns from the data. Unlike supervised learning algorithms, where tools like GridSearch can automate parameter optimization by systematically testing a range of parameter combinations against a performance metric, clustering algorithms often lack direct feedback mechanisms such as a clear accuracy measure. This is particularly true for algorithms used in unsupervised learning tasks like clustering, where the true labels are unknown, and evaluation metrics are less straightforward.

As a result, parameter tuning for clustering algorithms typically requires a manual or semi-manual approach. This involved a loop to test with various parameter combinations.

This study evaluates three clustering algorithms—BIRCH, DBSCAN, and GMM—to analyze student behavior based on log data from an educational platform. Each algorithm was meticulously tuned to optimize accuracy in detecting and interpreting behavioral patterns inherent in the logs. The parameters for each algorithm were carefully selected and adjusted through an iterative process involving multiple experiments with the student log data. This parameter tuning was essential to adapt each algorithm to the nuances of log-based data, which contained detailed records of student interactions. The specific tested tuning values of parameters are presented in Table 2, with the best parameters for each algorithm.

### 3) BIRCH

BIRCH (Balanced Iterative Reducing and Clustering using Hierarchies) is a clustering algorithm specifically designed

for larger datasets. Educational datasets typically feature varied data types and structures, reflecting students' diverse interactions and behaviors within a digital learning environment. BIRCH's incremental and dynamic clustering capabilities allow it to adapt effectively to such diversity. The algorithm processes data in small batches through its CF Tree (Clustering Feature Tree), which captures essential data summaries without the overhead of processing the full dataset at once. This method reduces the computational load and minimizes memory usage, crucial for handling larger datasets on limited resources [25].

Moreover, BIRCH's hierarchical nature is advantageous for educational data as it mirrors the layered complexity of learning behaviors and outcomes. By constructing a multi-level hierarchy of clusters, BIRCH can reveal subtle patterns and relationships within the data that simpler clustering techniques might overlook. This feature is particularly useful in educational settings for grouping students by learning behaviors, outcomes, or needs, providing actionable insights that can inform personalized learning interventions.

Finally, BIRCH's ability to refine and adjust clusters dynamically as new data becomes available supports ongoing monitoring and analysis of student performance and engagement. This is essential in educational settings where student behaviors and performance continuously evolve, requiring adaptive analytical approaches.




**FIGURE 9.** Visualization of BIRCH with PCA and UMAP.

Therefore, the BIRCH clustering technique was applied after refining the dataset's structure with PCA and UMAP. To suit the unique characteristics of the examined educational data, BIRCH's hyperparameters were fine-tuned, ensuring the best possible clustering given the size and nature of the data. The threshold for the BIRCH algorithm, which controls the maximum radius of the sub-clusters, was set to 0.3, providing a balance between the precision of the clusters and the algorithm's ability to scale. The branching factor, determining the number of child nodes in the BIRCH tree, was set to 70, allowing a comprehensive yet manageable tree structure. These hyperparameters were selected with an eye toward computational efficiency and the anticipated diversity within the educational data. Combining these settings with a predefined cluster count of four—based on the Elbow method and silhouette scores—it was possible to capture the data structure shown in Figure 9. It depicts the results of BIRCH clustering applied to the examined educational dataset, refined through PCA, and visualized using UMAP for two-dimensional embedding. The visualization distinguishes the following four clusters with varied sizes and characteristics:

- **Cluster 0** (Orange): Encompasses 37,996 data points, suggesting it contains the most common behavioral or characteristic patterns among the students.
- **Cluster 1** (Green): This cluster consists of 39,445 data points, making it almost as large as Cluster 0. This suggests the presence of another significant grouping with unique characteristics.
- **Cluster 2** (Red): A smaller cluster with 35,785 data points likely represents a specific subset of behaviors or academic engagement that differs from the larger clusters.
- **Cluster 3** (Blue): Includes 37,645 data points, indicating a substantial group with unique behavioral patterns or characteristics similar in size to Cluster 2.

The broad dispersion of points within Cluster 0 and Cluster 1 suggests that students in these groups engage in various activities within the course. They may participate in multiple interactions, from accessing study materials to completing assignments and quizzes. This variability indicates diverse behaviors and engagement strategies among these students.

In contrast, a more compact formation of Clusters 2 and Cluster 3 suggests that students in these groups might be engaging in more specific activities. Perhaps they were more focused on particular aspects of the course, such as specific types of assignments or content areas, indicating a more homogeneous interaction pattern.

#### 4) DBSCAN

As clustering techniques suitable for the educational dataset being examined are investigated, a transition is made from BIRCH to exploring the capabilities of DBSCAN. Unlike centroid-based methods, DBSCAN is adept at identifying clusters with irregular shapes and resilient to outliers, making it a robust choice for data with complex structures. DBSCAN operates on the premise of identifying core samples within high-density areas and categorizing non-core samples (which can either be 'border' points or 'noise') based on the density of their local neighborhood [25].

This approach is determined by two key hyperparameters: *min_samples* and *epsilon*. These parameters were iteratively refined through multiple experiments with the student log data to ensure a balance between the sensitivity of the algorithm to local point densities and its ability to distinguish between distinct clusters effectively. The DBSCAN

algorithm was finally configured with *min_samples* adjusted to 15 and *epsilon* set to 0.4 to identify substantive clusters while filtering out minor noise fluctuations. This setup allowed the algorithm to extend its reach sufficiently to include points in a neighborhood yet maintained tight enough clusters that accurately reflected genuine groupings in the data.

The resultant clustering, as shown in Figure 10, illustrates the application of DBSCAN to the refined educational dataset, visualized through the lens of UMAP for two-dimensional representation. This figure effectively demonstrates how DBSCAN articulates the complexities and intrinsic patterns within the educational environment, providing a clear view of the different student clusters.

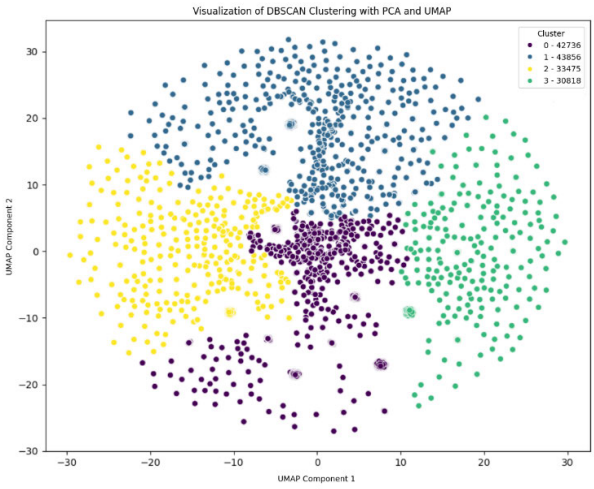



**FIGURE 10.** Visualization of DBSCAN with PCA and UMAP.

The visualization color-codes the following four distinct clusters, each representing a unique grouping within the student data:

- **Cluster 0** (Purple): Positioned predominantly in the lower part of the plot with 42,736 data points. This dense and concentrated distribution suggests a group of students consistently engaged in core course activities, showing a high uniformity in their interactions with structured course materials.
- **Cluster 1** (Blue): With 43,856 data points situated toward the upper part of the plot. This cluster is similar to Cluster 0 in terms of engaging in core activities. Still, it displays additional behaviors that suggest a blend of the consistent and exploratory approaches seen across other clusters.
- **Cluster 2** (Yellow) and **Cluster 3**(Green): These clusters, containing 33,475 and 30,818 data points respectively, are positioned on the left and right sides of the plot. Both clusters exhibit a wide range of activities, indicating a diverse and exploratory approach to coursework. Clusters 2 and 3 share similar engagement patterns, showing less concentration and more varied interactions compared to Clusters 0 and 1. Moreover, Clusters 2 and 3 are spread across the plot, suggesting that students in these clusters might be engaging in various course activities. This diversity could imply that these students were exploring a wide array of content or participating in different types of learning interactions that did not cluster around specific central tendencies. The broader dispersion might indicate less consistency in the type of activities students were engaged in, reflecting a more exploratory or varied approach to the coursework.

On the other hand, Clusters 0 and Cluster 1, located at the top and bottom of the plot, show a denser and more concentrated pattern of points. Such concentration suggests that the students in these clusters were likely engaged in more specific, perhaps even specialized, activities that were characteristic of their group. The density of points might indicate a higher consistency and uniformity in the activities they were engaged in, potentially signaling that these students were focused on particular aspects of the course that required similar patterns of interaction or study behaviors.

### 5) GMM

This comparative analysis also explores the use of GMM to uncover the latent structures within the educational dataset. GMM offers a probabilistic framework that accommodates the presence of subpopulations within the overall distribution. This aspect might be particularly relevant to educational data, which often contains such hidden stratifications. The GMM operates under the assumption that the data is generated from a mixture of several Gaussian distributions, each corresponding to a cluster. Each Gaussian, or 'component' in the mixture model has its own mean and covariance, which allows for acquiring a wide variety of cluster shapes and orientations that could be present in complex educational data [25].

The following parameters were set for GMM configuration through an iterative loop process to ensure the most effective clustering analysis. The optimal number of Gaussian components was determined to be 4. This decision was made to maintain consistency with the number of clusters identified in other clustering techniques used in this study, such as BIRCH and DBSCAN, allowing for comparability across different algorithms. To adequately capture the diverse patterns and relationships among data points, the 'full' *covariance* type was selected for GMM. This choice allowed each component to independently model its variance structure, accommodating the complexity and subtleties inherent in examined data. The maximum number of iterations was set to 250. This parameter ensured that the GMM algorithm had sufficient opportunity to converge on the best solution, reflecting an optimal balance between computational efficiency and accuracy in clustering analysis.

The GMM clustering results were visualized using a UMAP-reduced scatter plot (Figure 11). This visualization method enriches the understanding of the cluster by representing the data's underlying probability distributions. Each

data point is colored according to its highest posterior probability of cluster membership, offering a detailed illustration of the potential overlaps and ambiguities within the clusters. This nuanced approach highlights the probabilistic nature of GMM and provides a more complex view of student groupings compared to the definitive classifications typical of methods like BIRCH. The GMM clustering applied to the dataset and visualized using PCA and UMAP (Figure 11) effectively grouped the students into four distinct clusters, similar to the results obtained with BIRCH and DBSCAN.

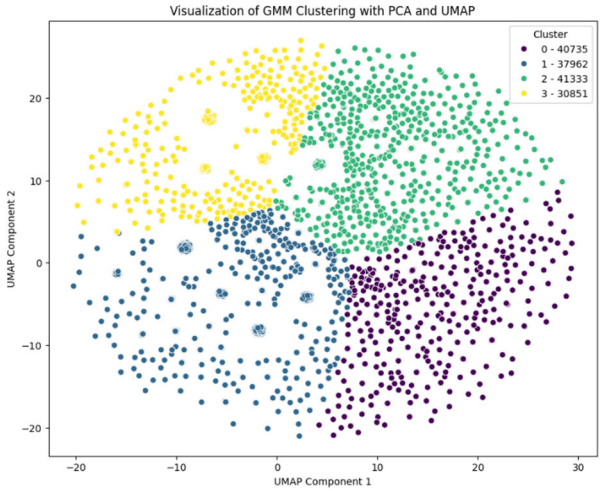



**FIGURE 11.** Visualization of GMM with PCA and UMAP.

The cluster distribution and engagement patterns revealed by GMM are as follows:

- **Cluster 0**(Purple): This cluster comprises 40,735 data points and is located primarily in the lower right part of the plot. It displays a variety of student activities, suggesting that these students were engaged in a broad range of coursework-related interactions.
- **Cluster 1**(Blue):This cluster, with 37,962 data points, is found toward the bottom of the plot. It is characterized by more specific, perhaps specialized activities, indicating that these students were likely engaged in distinct tasks that required similar patterns of interaction.
- **Cluster 2**(Green): This cluster contains 41,333 data points and is situated on the right side of the plot, merging slightly into the upper region. Similar to Cluster 0, it shows a wide range of activities, indicating diverse and exploratory approaches to the educational content.
- **Cluster 3**(Yellow): Including 30,851 data points positioned on the left side, this cluster indicates a group of students engaged in distinct, likely specialized activities similar to those observed in Cluster 1 but possibly involving different types of learning interactions or content areas.

### *E. EVALUATION*

In the previous phase, various clustering algorithms, such as BIRCH, DBSCAN, and GMM, were employed to analyze the student engagement data. The data were consistently formed into four clusters, with each method reinforcing the robustness of these groupings.

Each visualization and clustering technique successfully highlighted two distinct clusters of students, differentiated by their engagement levels and variety of activities. These clusters represent students with a wide range of interactions, including participation in quizzes, assignments, discussions, and other course-related tasks. The broad spectrum of activities within these clusters suggests that students exhibit diverse learning strategies and study habits, indicating variability in how they interact with the course content. This differentiation may reflect flexible engagement patterns, where some students explore multiple facets of the learning environment, while others may focus more narrowly on specific tasks.

The other two clusters identified in each model exhibited more specific and unique engagement patterns. These clusters consist of students with more narrowly focused activities, suggesting a specialization or a concentrated interest in particular aspects of the course. This type of clustering points to a group of students who might be focusing on specific tasks, projects, or learning outcomes, demonstrating a more targeted approach to their education.

In this evaluation phase, the quality and accuracy of the clustering results were assessed and validated. This phase serves as a critical measure to determine which clustering algorithm demonstrates the most effective performance using the input data from the study. Evaluation of clustering is an independent process distinctly separated from the clustering operation itself. It is conducted only after the clustering algorithms have produced their final outputs.

Two methods are practiced in measuring the quality of clustering results: internal and external validation. Internal validation is the process of evaluating clustering and comparing it to the results of the clustering itself, namely the relationship between the structures of clusters that have been formed. This is more realistic and efficient in solving problems involving educational datasets with increasing dimensions. This study used the following three types of internal validation methods that are often used in clustering studies:

- Davies-Bouldin index (DB),
- the silhouette coefficient index,
- Calinski-Harabasz index (CH).

Analyzing the performance metrics of three distinct clustering algorithms — DBSCAN, BIRCH, and GMM — on the educational dataset provides clear insights into their clustering efficacy. As a result, BIRCH stands out with the highest Calinski-Harabasz Index at 122,309, signaling clusters with excellent density and separation. This is complemented by its Davies-Bouldin Index of 0.769, the lowest among the three, which indicates that the clusters are not only well-separated but also less dispersed. The algorithm's superiority is further solidified by the highest Silhouette Score of

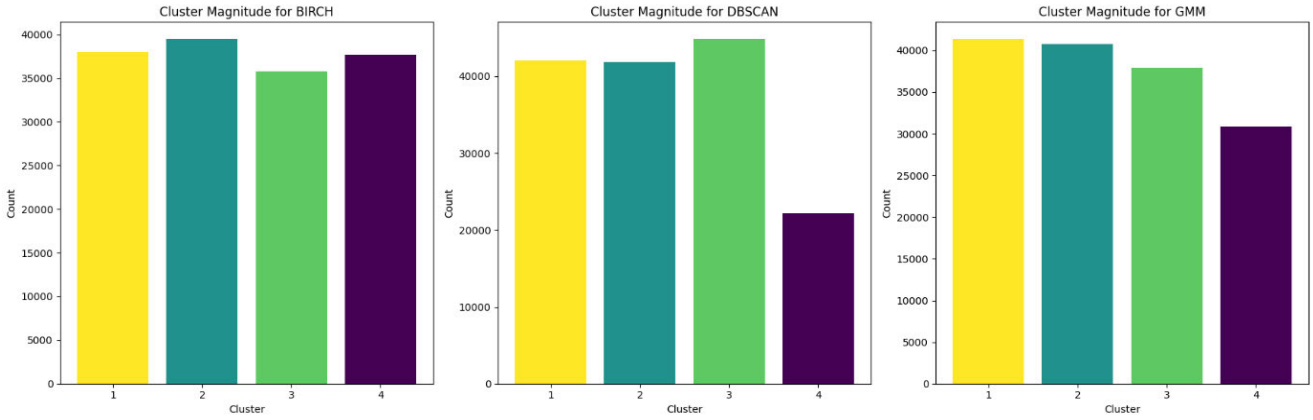



**FIGURE 12.** Comparison of cluster magnitudes for a) BIRCH, b) DBSCAN, and c) GMM.

0.565, confirming that BIRCH creates the most distinct and well-defined clusters among the algorithms tested (Table 3 ).

**TABLE 3.** Results of parameter tunning.

| Validation Metrics | BIRCH | DBSCAN | GMM |
|---|---|---|---|
| *Davies-Bouldin Index* | 0.769 | 0.966 | 0.832 |
| *The Silhouette Coefficient Index* | 0.565 | 0.328 | 0.435 |
| *Calinski-Harabasz Index* | 122 309 | 94 906 | 119 669 |

GMM follows closely with a Calinski-Harabasz Index of 119,669, suggesting that it also forms well-defined clusters, albeit slightly less so than BIRCH. Its Davies-Bouldin Index, while low at 0.832, is not quite as optimal as BIRCH's, pointing towards slightly more dispersed clusters. While the Silhouette Score of 0.435 for the GMM algorithm indicates moderate performance, it suggests that the clusters are relatively distinct and separated, though not as robustly defined compared to BIRCH.

Lastly, DBSCAN, with a Calinski-Harabasz Index of 94,906, fell behind the other two methods, suggesting its clusters are less dense and more overlapping. This interpretation is supported by the Davies-Bouldin Index of 0.966, the highest value reported, which further implies that the clusters are less compact and more intermingled. The Silhouette Score for DBSCAN is 0.328, which suggests that the clusters it identifies are reasonably distinct but may not exhibit as strong cohesion or separation as those formed by other algorithms such as BIRCH and GMM. Simultaneously, this score indicates that while the clusters are identifiable and separated from each other to a certain degree, they are tightly grouped and may showcase an overlapping cluster structure.

Given the context of Moodle LMS logs, where a balanced and clear segmentation of student behavior is crucial, BIRCH proves to be the most effective algorithm. It not only provides the most balanced cluster distribution but also maintains high separation and low dispersion, making it the best fit for such data. While GMM shows promise, particularly if probabilistic cluster memberships are of interest, BIRCH's ability to create distinct and evenly distributed clusters makes it the preferred choice for application. In conclusion, BIRCH's performance across all metrics indicates it is the superior algorithm for clustering in this context. It offers a robust solution for discerning patterns in student engagement and learning behaviors within Moodle.

In addition to assessing the internal validity of the obtained clustering results through metrics, the magnitude of each cluster was also examined. By visualizing the number of data points in each cluster, the relative size and distribution of clusters could be observed, offering a clearer picture of the overall clustering landscape. This approach helps in understanding the scalability of each cluster and in identifying any potential imbalances or biases in how data points are grouped, which is crucial for interpreting the practical significance of clustering outcomes.

The BIRCH magnitude results (Figure 12a) demonstrate a relatively balanced distribution of data points across four clusters, with cluster sizes ranging closely together, indicating a uniform dataset segmentation. Figure 12b displays a different pattern, where one of the clusters (Cluster 3) is significantly smaller than the others. This could suggest that DBSCAN identified a denser but smaller cluster, possibly an outlier group or a subgroup within the dataset. The GMM magnitude results (Figure 12c) indicate that the GMM partitioned the data into four clusters with varying sizes. Similar to BIRCH, the GMM produced more balanced clusters. However, the size of Cluster 4 is noticeably lower than the others, which might reflect a subgroup with distinct characteristics or a higher degree of separation from the rest.

As a result, it can be confirmed that each algorithm's clustering reflects its underlying principles and the nuances it detects in the data. BIRCH's even distribution suggests that uniform groupings were found across its clustering features. DBSCAN's varied cluster sizes indicate its density-based

approach, potentially revealing less dense areas or outliers as separate clusters. GMM's result, with one smaller cluster, suggests the presence of a distinct group with a specific probability distribution that differs from the larger clusters. These bar charts not only provide a quantitative comparison of cluster sizes but also highlight the differences in how each algorithm interprets the same data. Such insights can offer the choice of the most suitable algorithm for particular types of analysis or encourage further investigation into the distinct characteristics that define each cluster.

Considering the computational resource usage and performance metrics alongside the previously discussed clustering quality metrics gives a comprehensive understanding of the efficiency and effectiveness of the algorithms applied to the educational dataset. This analysis was conducted on a laptop equipped with an Intel Core i7 9750H Coffee Lake processor, a configuration that includes 16GB DDR4 RAM and dual storage options with a 512GB SSD for fast access and a 1TB HDD for additional capacity. The laptop also features an NVIDIA GeForce RTX 2070 Max-Q graphics card, ensuring processing power for complex data operations. To assess the performance and resource utilization of different clustering algorithms, results were averaged from 150 tests, providing a solid statistical base for comparison. This setup allowed for a detailed examination of each algorithm's CPU and memory usage under controlled conditions.

DBSCAN exhibited a higher CPU usage at 25.67%, which is considerably more than the other two algorithms. This is likely due to the complexity of dealing with a density-based clustering approach. However, its memory usage was moderate at 1.10%, and it performed its clustering relatively quickly, taking just 4.66 seconds to complete.

BIRCH demonstrated excellent efficiency with the lowest CPU usage at 7.51%. Its memory usage was also slightly lower than DBSCAN's at 1.01%, suggesting that BIRCH is resource-efficient. Despite taking the longest time to run at 5.55 seconds, considering the high quality of its clustering indicated by the earlier metrics, the slightly longer runtime may be a justifiable trade-off.

GMM strikes a balance between resource usage and performance time. It required more CPU resources than BIRCH (9.55%) but less than DBSCAN (1.15%), and it used slightly more memory than the other two algorithms (1.15%). Notably, GMM was the fastest, completing its task in just 2.76 seconds. This indicates a high level of computational efficiency, particularly when the clustering quality is also considered.

In summary, if the primary concern is the quality of clustering, BIRCH appears to be the superior choice due to its exceptional performance in creating distinct and well-separated clusters. However, GMM stands out as the most computationally efficient algorithm, with a balance of lower CPU usage and the fastest processing time. Although DBSCAN utilized more CPU resources, it maintained reasonable memory usage and performed clustering tasks rapidly. The decision between these algorithms may ultimately hinge on the specific requirements of the use case. If computational resources are limited, BIRCH might be favored for its lower CPU and memory usage. On the other hand, if speed is the priority, GMM's quick processing time presents a compelling case, especially since it also offers high-quality clustering results.

### F. DEPLOYMENT

In adapting the deployment phase of the presented CRISP-DM methodology, the focus shifts from traditional model deployment to validating and interpreting the relevance of the clustering outcomes within educational settings. This step is crucial as it allows to understand and effectively apply the insights gained from the clustering process to real-world educational scenarios.

This research section focuses on the temporal dynamics of student interactions with the course during the semester. However, before the examination of the specific characteristics of the input attributes, it is important to note the scale of the used dataset. The dataset contained a total of 150,881 records from 152 students. This extensive dataset means that the clusters identified are not unique to individual students; a single student may appear across multiple clusters. This overlap highlights the complex nature of student behaviors and interactions with the course content, demanding a nuanced approach to analyzing cluster characteristics and their implications.

Initially, the attribute *weekday* was examined, which represented the days of the week—Monday through Sunday—during which students accessed the course. Examining the distribution of course logins across different weekdays, patterns of engagement and peak activity days can be inferred. This analysis is visualized in Figure 13, which displays the frequency of course access for each day of the week. By examining these patterns, deeper insights can be gained into how students interact with the learning management system at different times, which can be pivotal for understanding their engagement and learning behaviors.

In the context of this analysis, it is crucial to consider the course schedule to fully understand the temporal dynamics of student interactions. Exercises for the course were scheduled on Mondays, Tuesdays, Thursdays, and Fridays. However, analyzing exact student attendance on these specific days may not provide meaningful insights due to the flexibility in attendance. Students often attended sessions on the most convenient days rather than strictly adhering to the prescribed schedule. This flexibility in attendance patterns needed to be considered as it may influence the typical engagement metrics and could obscure the impact of scheduled activities on actual student engagement.

For students in Cluster 0 and Cluster 1, the data shows a relatively stable access pattern with peak activity on Tuesday, Thursday, and Friday, Saturday. This suggests that students in this cluster engage diligently with course materials, aligning their study efforts with the structured schedule of the course exercises held on Mondays, Tuesdays, Thursdays, and

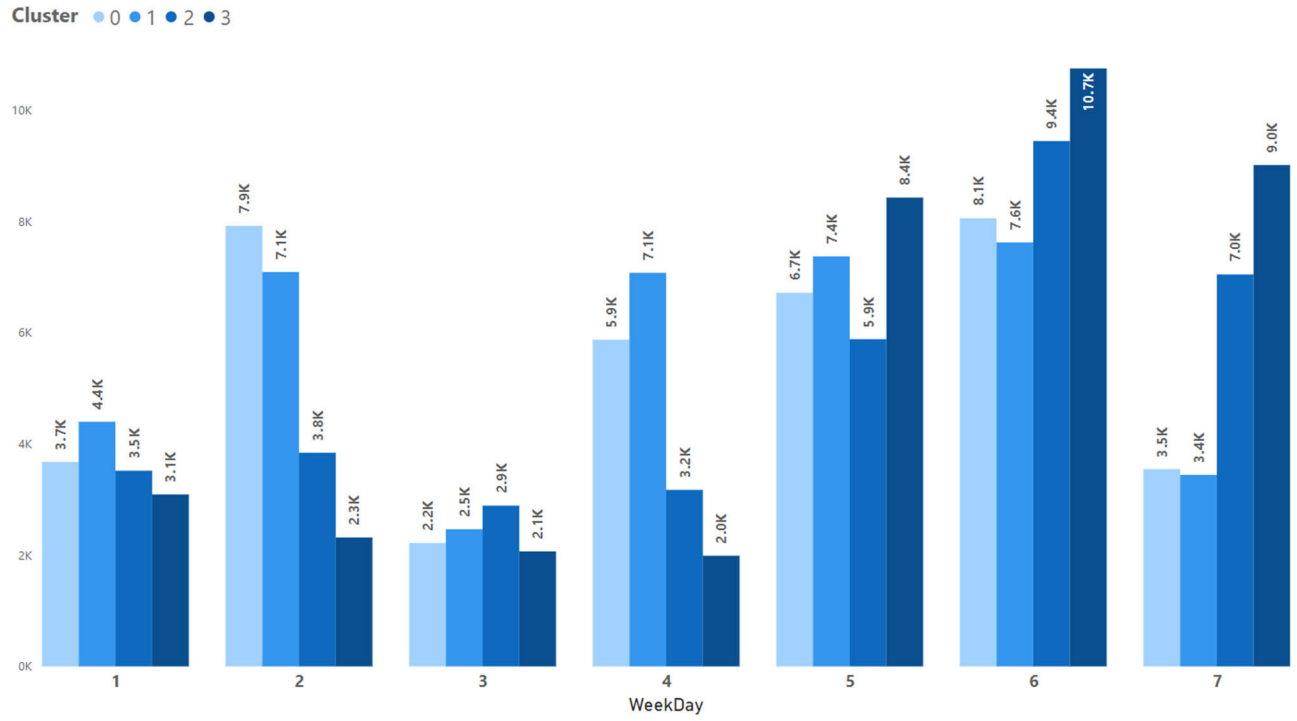



**FIGURE 13.** Student behavioral analysis by attribute weekday for different clusters.

Fridays. The continued high activity on Friday and Saturday may indicate that these students not only finish any pending duties but also possibly use the weekend to get ahead in preparation for the upcoming week. This pattern suggests a proactive approach to managing their coursework, where weekends are utilized to finalize pending assignments and study in advance. This behavior reflects a diligent and responsible attitude toward academic commitments.

Cluster 2 displays a distinct pattern of increased activity over the weekend, with Saturday showing the most significant engagement. This trend may suggest that students in this cluster face commitments during the weekdays, such as work or family responsibilities, that limit their weekday course interactions. Alternatively, they might prefer to concentrate their study efforts over the weekend. It is also possible that these students had scheduled course exercises on Friday, which could lead to lower activity levels during the rest of the week, as they might use weekdays to prepare for these sessions and the weekend to complete associated tasks or further study the material.

Cluster 3 again has very low weekday activity, with a sharp increase on Saturdays. This pronounced weekend activity could suggest that students defer most of their coursework to the weekend, potentially indicating possible procrastination, time management challenges, or other external constraints during the weekdays.

Activity levels are noticeably lower on Wednesday in each cluster, aligning with the absence of scheduled course exercises. This dip is consistent with the overall pattern observed across the dataset, where Wednesday looks like a break in the academic week. The reduced engagement on this day does not necessarily signify a lack of responsibility but rather a regular decrease in activity, which can be utilized for other duties.

As can be seen, distinct engagement patterns across the clusters have been identified from this analysis. Clusters 0 and 1 demonstrate relatively active engagement throughout the weekdays, indicating a consistent involvement with the course materials during the scheduled days. Cluster 2, although less active during weekdays, may not necessarily reflect disengagement; this pattern could be attributed to students primarily enrolled in Friday sessions as per the course schedule. Lastly, Cluster 3 significantly differs from the others, showing the lowest overall activity, which might suggest different challenges or priorities that affect their course participation.

Given these insights, the decision was made to realize a further analysis of weekly engagement throughout the semester (Figure 14). This next phase could help to understand how consistently students participate in the course over time and may reveal additional patterns or needs specific to each cluster.

Cluster 0's semester-long activity pattern begins with gradual engagement, likely reflecting an initial orientation period and adjustment to the course. As the semester progressed, there was a notable increase in activity, peaking around the mid-semester mark, possibly due to the buildup of coursework and preparation for mandatory assessments. The sharp decline in activity during Week 6 is aligned with a scheduled

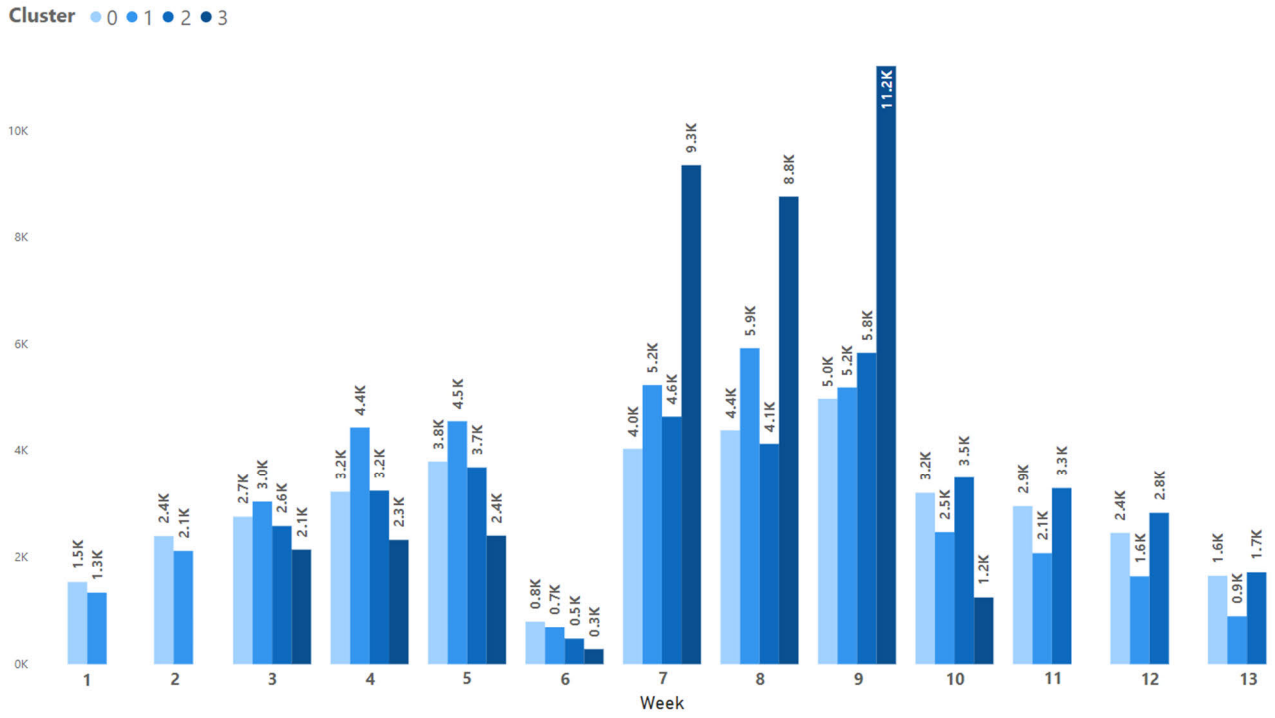



**FIGURE 14.** Student behavioral analysis by attribute week.

holiday, leading to a predictable period of reduced activity as students take time off. Then, there was a sudden increase in activity, suggesting a period of concentrated effort as students returned to their studies and began preparing for important evaluations. The pronounced peak in Week 9 corresponds with the schedule of a mid-term test, likely driving a surge in course logins as students access study materials and review sessions offered through the LMS. The spike in activity demonstrates the students' responsiveness to impending assessments and their commitment to preparation. After Week 9, the subsequent decrease in logins may be attributed to relief after the mid-term test, with students potentially focusing on other courses. As the semester draws to a finish, the activity lessened further, reflecting the end of regular classes and a possible shift to offline study methods as final assessments approach.

Therefore, Cluster 0 represents a cohort of students whose engagement was significantly influenced by the course timeline, displaying flexibility and a reactive approach to the academic workload and assessment schedule. The level of activity during the initial weeks of the semester is significantly higher compared to other clusters. The objective of these students appeared to be the mid-term examination, which their efforts throughout the semester were leading up to. The decrease in accesses following the mid-term test is less significant than for the other clusters. This suggests that the students were more responsible for their course activities.

Cluster 1 exhibits a pattern of engagement over the semester that is remarkably similar to the aforementioned Cluster 0, with a gradual increase in activity, peaking around key academic weeks. Notably, the initial activity in the first two weeks was low, indicating a slower start or less immediate engagement with the course. However, unlike Cluster 0, the activity in Cluster 1 significantly declined in the final weeks of the semester, suggesting that students may shift their study efforts away from the LMS to other methods of preparation for final exams or completing assignments. During the mid-semester weeks, Cluster 1 shows higher activity levels. This increase could be influenced by a larger number of students, naturally resulting in more logins and interactions within the LMS. Despite this, the consistency in engagement during these weeks indicates a stable pattern of accessing course materials, likely in response to upcoming assessments or deadlines.

Cluster 2 indicates an activity beginning from Week 3, suggesting that these students missed crucial early coursework, possibly affecting their foundational understanding and preparation for subsequent material. The activity remained modest but steady through the middle of the semester, which could imply a consistent yet minimal interaction with the course materials or platform. This pattern suggests that while these students participated, they might be less reliant on the LMS or possibly balancing their coursework with other commitments. The peak around Week 9, which coincides with an exam period, indicates a surge in logins, perhaps as students seek to access revision materials, past papers, or other resources in preparation for exams. The rise in activity around an exam period is a common trend, often reflecting

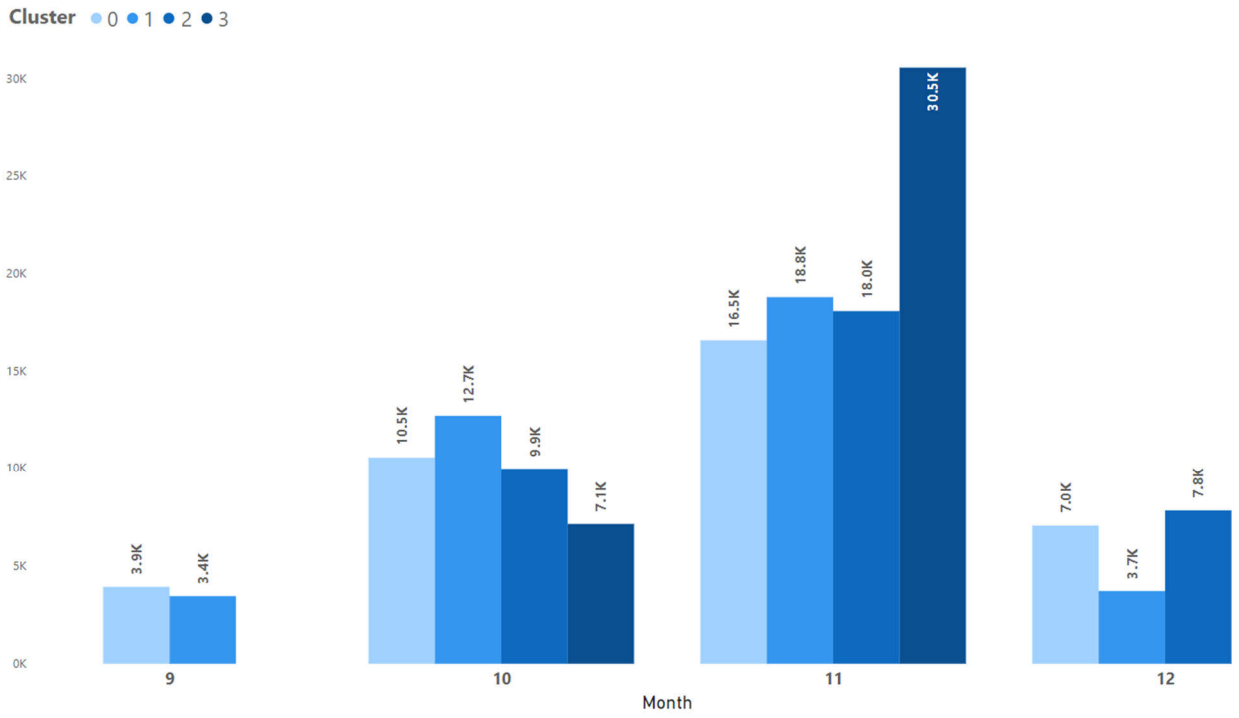



**FIGURE 15.** Student behavioral analysis by attribute month.

last-minute studying or increased use of practice quizzes and other revision tools available on the LMS. Contrary to common assumptions, there is a noticeable continuation of high activity in the LMS after the exam period. This sustained interaction might suggest that students took advantage of opportunities to improve their grades during this period. After the exams, they got the chance to retake the mid-term test, allowing students to enhance their overall performance. The continued engagement could also indicate that students utilized this time to complete or resubmit coursework, thereby seeking to maximize their academic outcomes before the end of the term.

On the contrary, Cluster 3 indicates sporadic and low engagement throughout the semester, with a noticeable peak in activity around Week 9. This is a unique pattern compared to the other clusters. In early Weeks (1-3), there appears to be no activity, which could indicate severe issues with engagement or a misunderstanding about the course requirements, particularly in a blended learning environment where early interaction is crucial. During weeks 4-7, the engagement was still minimal. It did not show a strong upward trend, suggesting that the students in Cluster 3 may not fully utilize the online resources provided or may be relying more heavily on the in-person aspects of the course. The noticeable increase in activity starting Week 7 and peaking in Week 9 indicates a last-minute surge in LMS usage, likely due to students attempting to catch up and prepare for the mid-term test. This spike reflects an intensive, albeit brief, engagement period where students may have been trying to compensate for earlier disengagement. After the mid-term test, there is a sharp drop in activity, with engagement levels tumbling down further in Week 10 and becoming almost non-existent in Weeks 11 to 13. This pattern suggests that the peak was an isolated event, as students returned to their previous minimal usage of the online course components after the test. This demonstrates that no efforts to enhance continuous involvement were effectively maintained.

The engagement patterns observed across the different clusters during the semester reveal consistent behaviors both weekly (Figure 14) and monthly (Figure 15), reflecting the interconnected nature of academic timelines. Clusters 0 and 1 demonstrate very similar and stable patterns of interaction with the LMS, maintaining steady activity levels throughout the various weeks and months. This consistency suggests that students in these clusters regularly engaged with the course content, aligning their study habits closely with the academic calendar.

In contrast, Cluster 2 shows a unique engagement. There was zero activity in the first month, followed by average levels in October and November. Notably, the activity is higher in December compared to other clusters. This suggests that despite potential time constraints earlier in the semester, students in Cluster 2 might utilize the final month to catch up on or complete tasks or perhaps to improve their overall course grades. This spike in activity could be driven by the closing of grading periods, prompting students to intensify their use of the LMS. Overall, this pattern suggests that students in Cluster 2 might be engaging with the LMS sporadically, possibly

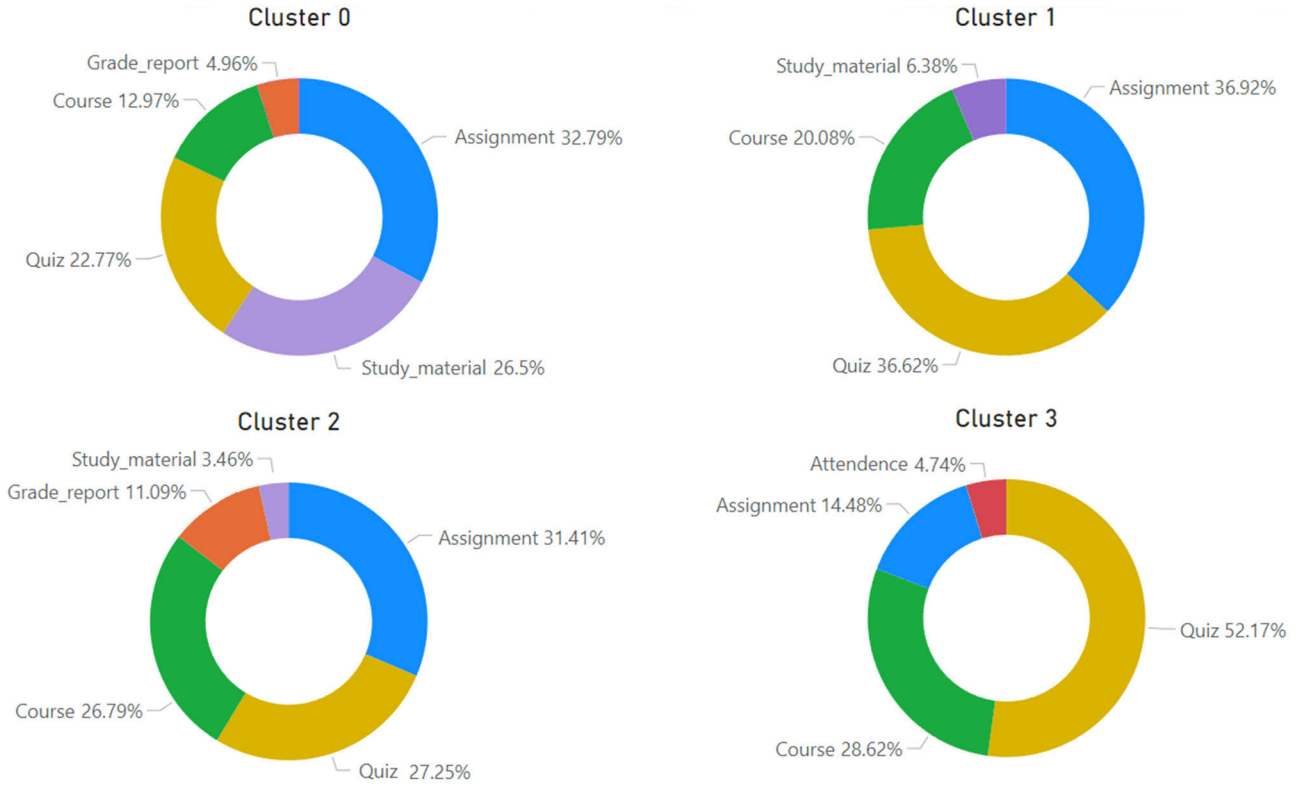



**FIGURE 16.** Student behavioral analysis by attribute component.

intensifying their use around critical academic periods, such as mid-term tests. Maintaining engagement through periodic check-ins or reminders during weeks of lower activity could help sustain their academic involvement.

The last Cluster 3 exhibits no activity in September and December, with the lowest engagement occurring in October. All significant interactions were concentrated around November, likely in response to mid-term assessments. This spike in activity during a single month suggests that students in this cluster might only engage with the LMS when critically necessary, possibly due to the mid-term test. Supporting these students effectively might require targeted interventions designed to enhance engagement, ensuring that they are not only preparing for tests but also consistently engaging throughout the semester.

Beyond analyzing when students access the course, examining what specific course components they were engaged with is equally important. This approach helps to understand the temporal engagement patterns and the content that attracts the most attention or is possibly the most challenging. By identifying which resources were most frequently accessed or which activities reported the highest participation, educators can gain valuable insights into student preferences and potential areas of difficulty within the course.

Figure 16 illustrates how students interacted with various course elements in different clusters. Although there were no mandatory assignments, the value of 32.79% for Cluster 0 suggests a high engagement with optional tasks or activities that students chose to undertake, suggesting a proactive approach to their learning. The engagement with study materials (26.5%) also remained significant, indicating that students frequently utilized provided resources such as lecture notes and other resources to aid their learning. This underscores their dedication to understanding the course content thoroughly. The quiz component, with a value of 22.77%, includes various quizzes under thematic units in addition to the mid-term test, showing a robust engagement. This suggests that quizzes served as both reinforcement of learning and as a significant measure of student self-assessment throughout the course. The course content (12.97%) category includes direct interactions with the core content, such as instructional videos or interactive modules. The relatively lower engagement could suggest that students might prioritize active learning tools (like quizzes and assignments) over passive content consumption. Despite being relatively low, the engagement with grade reports (4.96%) observed in Cluster 0 reflects a targeted interest in this component among students. This suggests that students in this cluster might be strategically checking their grade reports to identify opportunities for improving their marks towards the end of the semester, particularly once other coursework has been substantially completed.

A balanced distribution across various course components was observed in Cluster 0. On the other hand, Cluster 1 shows

a more concentrated focus on assignments (36.92%) and quizzes (36.62%), which together dominate the engagement landscape. This cluster's students might probably prioritize activities directly tied to assessment and immediate feedback.

Engagement with course content (20.08%) was substantial but less than that seen with assignments and quizzes, indicating a preference for direct application and assessment over broader content exploration. Study materials were the least engaged (6.38%), suggesting that students might rely more on core materials and direct assessments to guide their learning.

Based on the previous assumptions and the information gathered, Cluster 2's primary periods of activity were concentrated in the weeks leading up to and following the mid-term test. This pattern of engagement is reflected in the components they interacted with most frequently. The high levels of activity in assignments (31.41%) and quizzes (27.25%) suggest that students were likely attempting to absorb as much knowledge as possible from these resources. Such engagement indicates a strategic effort to prepare thoroughly for the mid-term assessments and deeply understand the course content (26.79%). The sustained activity observed after Week 9 through the end of the semester likely relates to students' endeavors to improve their grades after the mid-term test. This prolonged engagement suggests that students continued to leverage the LMS extensively to enhance their understanding and possibly to take advantage of opportunities for grade improvement. Whether through additional quizzes, revisiting course materials, or completing supplementary assignments, their consistent use of the LMS after mid-term test indicates a proactive approach to maximizing their academic outcomes. While Cluster 2 demonstrates significant interaction with assignments, quizzes, and course content, it is notable that their engagement with supplementary study materials was markedly lower. This limited interaction with study materials, accounting for only 3.46% of their overall engagement, suggests that these resources were not considered critical to their learning strategy. This behavior underscores a keen student's interest in passing their courses using available educational resources. Such insights into engagement patterns highlight the importance of providing continuous access to learning aids and support throughout the semester, catering specifically to the needs of students who intensify their study efforts, at least during crucial academic periods.

The engagement in Cluster 3 underscores a highly focused approach, predominantly around quiz participation (52.17%), which far exceeds other components. This dominant focus on quizzes supports the hypothesis that students in this cluster prioritized direct testing opportunities around the mid-term week, possibly as a mean to gauge or enhance their understanding rapidly. Course content (28.62%) also plays a significant role, aligning with the need for immediate access to relevant information to their quiz preparations. Conversely, the low engagement with assignments (14.48%) and checking up on their attendance (4.74%) suggests a selective interaction with the course, driven by the necessity of assessment rather than a comprehensive educational experience.

Following the comprehensive evaluation of student engagement across the clusters, the final step is to delve into the specific actions that students have undertaken on various course components (Figure 17). This analysis could provide a deeper understanding of how students interact with the LMS and which activities on components dominate their engagement.

Cluster 0 shows a balanced distribution of actions, with the majority being viewing (51.64%) and downloading (32.52%) materials and a significant portion (15.84%) of activities related to submission, which likely includes submitting assignments and completing quizzes. This suggests an active and engaged learning approach, with students taking advantage of the LMS to access and submit course materials.

Cluster 1 has a higher proportion of viewing (68.35%) and a smaller amount of downloading (7.29%) and submitting (24.36%). The emphasis on viewing over downloading could indicate a preference for online engagement with content directly through the LMS rather than working with offline materials. The relatively high submission rate shows that these students also actively complete and submit assignments or assessments.

Cluster 2 is dominated by viewing activities (80.7%), with very little downloading (2.19%) and a moderate submission rate (17.11%). The overwhelming focus on viewing might suggest that students used the LMS primarily as a reference or review tool, possibly for going over lectures and notes, with less emphasis on submitting assignments.

Cluster 3 also primarily engages in viewing activities (75.83%) but has a higher submission rate (23.59%) compared to Cluster 2 and almost negligible downloading (0.5%). The substantial quiz submission activity (52.17%) is closely tied to previous findings of learning strategy as students prepared for their mid-term examinations. This pattern indicates a focused effort where students were intensively engaged in quizzes to absorb and review course material rapidly. This indicates a group of students who were more focused on the consumption of information, with little to no material being downloaded for offline study.

The data across all clusters underscores the central role of the LMS as a viewing platform for course material, with varying degrees of interaction with downloadable content and assessment submission. For Clusters 2 and 3, strategies that encourage broader and more interactive use of the LMS features could be beneficial. For instance, incorporating interactive elements that cannot be downloaded, such as in-browser assessments, might increase overall engagement with the LMS and ensure a more holistic use of the platform's capabilities.

In this research, the decision not to focus on the variable *target* within the LMS logs was driven by a strategic choice to concentrate on variables that provide a broader understanding of student behaviors and interactions in the learning management system. The *target* variable is specific to particular activities or content within the course, which may not generalize well across different courses or disci-

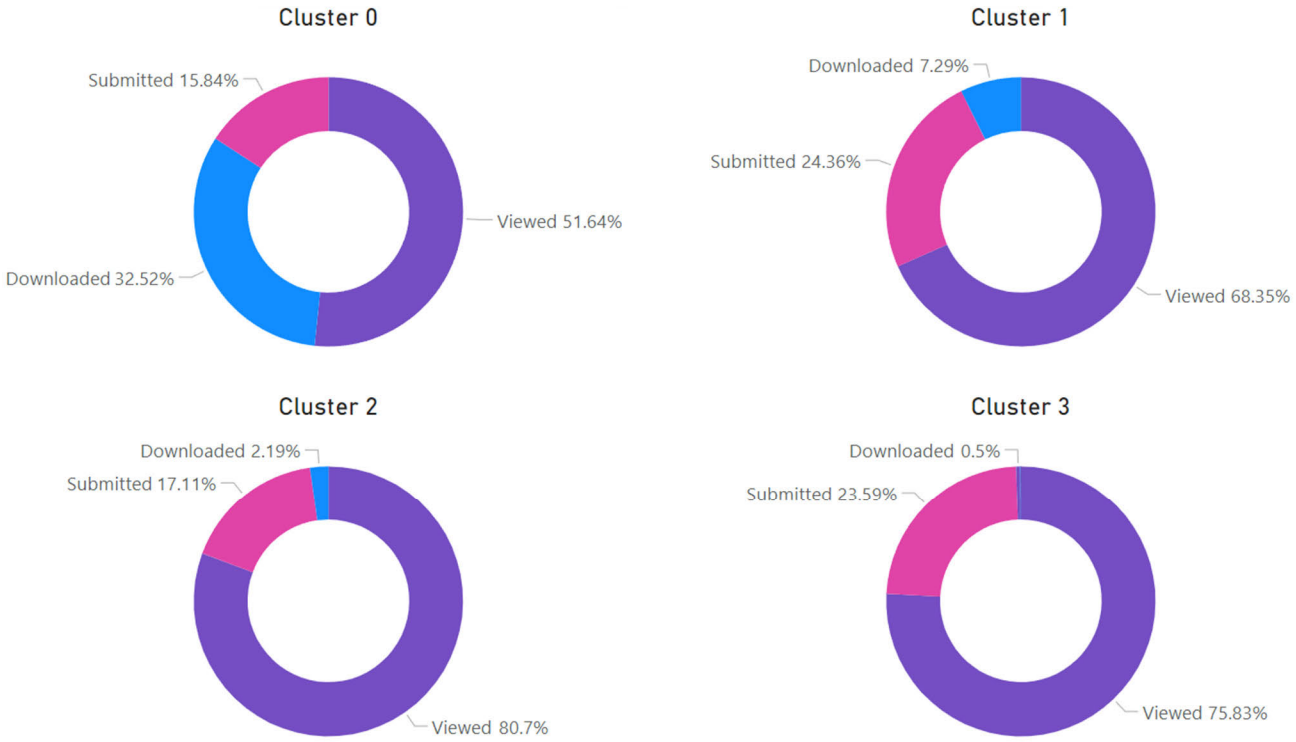



**FIGURE 17.** Student behavioral analysis by attribute action.

plines, potentially limiting the scope and applicability of the findings. By focusing on more generalizable variables like *component* and *action*, the analysis aimed to uncover more universally applicable insights across various educational contexts, thereby enhancing the robustness and relevance of this study.

## IV. DISCUSSION

Combining all the analyses of the clusters' time and activity patterns presents a multi-faceted view of student engagement in a blended learning environment. We will discuss it more precisely in this section from the segmentation and possible contribution to the student dropout prediction point of view.

### A. QUALITATIVE CHARACTERISTIC OF FOUND CLUSTERS OF STUDENTS BASED ON THEIR PERFORMANCE

Cluster 0 represents a group of students who took a comprehensive approach to the course. They engaged steadily across the semester and actively used the LMS for a wide range of activities. They showed a pattern of accessing, viewing, downloading materials, and submitting assignments, suggesting consistent engagement with both content consumption and assessment completion.

Cluster 1 exhibits high student engagement primarily through viewing materials on the LMS, which peaks around assessments, indicating their use of the platform as a direct learning tool. Despite lower downloading activities, their submission rates remained robust, suggesting a balance between online interaction and completing evaluative tasks. Similar to this pattern, the cluster maintains a high level of activity throughout the entire semester, reflecting a consistent reliance on the LMS for both learning and assessment preparation.

Cluster 2 stands out with a high focus on viewing content, especially around the midterm period, with very little content being downloaded. This emphasizes a pattern of involvement that is mostly influenced by immediate academic needs rather than consistent study habits. In addition, they thoroughly examine assignments, quizzes, and course materials before the mid-term examination, as seen by their engagement with grade reports. This behavior demonstrates their prioritization of academic achievement, particularly during crucial evaluation periods rather than consistently throughout the entire semester.

Cluster 3 is characterized by a low level of activity throughout the semester, with most interaction concentrated around mid-term tests, which represent a semi-mandatory activity. Their primary engagement revolved around quizzes, indicating their involvement in both viewing and submitting. However, they showed little interest in assignments and study materials, as evidenced by their minimal interaction with these components.

Overall, these analyses suggest diverse engagement strategies among the student clusters. While some demonstrate a balanced and sustained use of the LMS, others are more assessment-driven in their interaction. The insights gained underscore the importance of tailoring support to meet the varying needs of students and enhancing the effectiveness of the blended learning experience. Strategies such as interactive and mandatory online activities and early engagement

monitoring could be implemented to encourage a more uniform engagement across all clusters and throughout the semester.

To better understand the connection between student engagement and academic performance in a blended learning environment, this study has taken the next step of correlating the observed activities within each cluster to the actual academic outcomes of the students. By mapping grades and other measures of academic success onto the identified patterns of LMS usage, the aim is to expose how different modes of interaction with course materials may influence student performance. In this endeavor, the focus is on discerning whether certain behaviors – such as regularity of LMS usage, diversity of activity types, and the timing of interactions – can be linked to higher grades or improved comprehension of course content. An investigation of whether patterns of minimal engagement or last-minute activity spikes correlate with lower academic achievement is also done.

The outcome of this analysis is expected to provide educators with valuable insights that could shape course design, the scheduling of assessments, and the allocation of resources. Additionally, it could help identify which student behaviors should be encouraged or reformed to promote better academic results. By closely aligning the used educational strategies with the data-driven understanding of student engagement, the goal is to boost academic performance and enhance the overall learning experience within the blended courses.

The course timeline was divided into four equally spaced periods to better analyze student performance. Everything students did during the first three weeks of the course was included in the submissions attribute in the first quarter (Q1). In addition to the first quarter data, the second quarter (Q2) data included newly recorded interactions up to the mid-course. The same reasoning was followed in the third and fourth quarters (Q3 and Q4) (Table 4 ). Simultaneously, the student's achievements were mapped in these submissions to four ordinal values to normalize the differences between absolute values of received grades.

As a result, the following insights summarized in Table 5 were achieved, which have also been supplemented with time-series visualizations of individual attributes across the quartiles, as shown in Figure 18. Cluster 0 is characterized by a robust pattern of engagement with the LMS, where students participated actively and systematically throughout the semester. Their interaction with the LMS was not limited to just accessing study materials but also involved regular submissions and high achievement in quizzes and mid-term test. This persistent approach has yielded positive results in their academic performance, with a majority regularly achieving higher grades and maintaining a 100% pass rate, indicating no dropouts. This group's consistent and disciplined interaction with the LMS, including a balanced approach to viewing, downloading, and submitting assignments, correlates with their strong academic outcomes, mostly ranging from A to C grades.

**TABLE 4. List of performance attributes with description.**

| No. | Attribute | Values | Data Type |
|---|---|---|---|
| *1* | Submissions Q1 | None<br>1<br>2<br>above 3 | Ordinal |
| *2* | Submissions Q2 | None<br>1<br>2<br>above 3 | Ordinal |
| *3* | Submissions Q3 | None<br>1<br>2<br>above 3 | Ordinal |
| *4* | Submissions Q4 | None<br>1<br>2<br>above 3 | Ordinal |
| *5* | Quizzes | below 6.00<br>6.00 – 7.00<br>7.00 – 8.00<br>8.00 – 10.00 | Ordinal |
| *6* | Mid-term test | below 9.00<br>9.00 – 11.00<br>11.00 – 13.00<br>13.00 – 15.00 | Ordinal |
| *8* | Grade | A, B, C, D, E, Fx | Ordinal |
| *9* | Dropout status | Pass (1)<br>Fail (0) | Nominal |

Cluster 1 exhibits patterns comparable to Cluster 0, demonstrating consistent engagement and an even greater frequency of participation in the LMS. Regarding academic performance, it has been observed that the assignments they actively worked on were submitted quite effectively. However, regarding the quizzes, the outcomes were somewhat inferior compared to Cluster 0. This can be linked to the cluster's lack of emphasis on study materials, which is also seen in the subsequent grades, where around 79.41% of the grades were D's. However, the overall success rate was 100%.

Cluster 2 started the semester with sparse engagement, possibly missing crucial early interactions with the course content on the LMS. Over time, students' activity increased, specifically around periods with significant academic events such as mid-term test, suggesting a reactive approach to study habits driven by assessment timelines. During the mid-term exam period, there was a noticeable rise in students' activity. However, this increase did not lead to outstanding performance in submissions in Q3 and Q4.

Although they were actively involved, most students did not achieve the highest score ranges. The assessment of quizzes and the mid-term test yielded a low outcome, with

most students in this group achieving an E grade, accounting for 85.35%. In addition, there was a dropout rate of 14.65%, underscoring the impact of irregular study habits on academic achievement.

Cluster 3 demonstrated minimal engagement, with an enormous peak during the mid-term test period. The limited interaction with the LMS throughout the semester and the low academic performance, particularly with the majority scoring the lowest grades (Fx), suggest that students in this cluster might have struggled to engage effectively with the course. Their peak activity during the mid-term test period was insufficient to elevate their overall academic performance, as evidenced by their dropout rate (61.54%) and lower grades. Perhaps it could have been possible to mitigate this high dropout rate if students had utilized the final weeks to improve their grades or exerted effort similarly to what was observed in Cluster 2. Overall, the analysis across all clusters reveals a clear trend of consistent and varied engagement with the LMS correlates with better academic outcomes.

Cluster 0 exemplifies how systematic use of all available resources and regular submission of assignments can lead to high academic achievement. Meanwhile, the sporadic and lower engagement levels in Clusters 2 and 3 underscore the need for strategies to enhance consistent interaction with the course content throughout the semester. The insights suggest that encouraging regular engagement with the LMS and providing support mechanisms for students to adopt efficient study habits could lead to improved academic performance and reduced dropout rates. These findings highlight the importance of creating interventions that align with students' diverse needs and engagement patterns to enhance their academic success in a blended learning environment.

### B. CLASSIFICATION MODEL OF STUDENTS EARLY DROPOUT BASED ON RESULTS OF STUDENTS' PERFORMANCE CLUSTERING

Given that the log files stored within the LMS Moodle have proven to be beneficial in understanding students' behavior within the system and strongly correlate with their academic performance, it was decided to utilize classification models to assign these records to identified clusters. Predictions for specific periods have been made as it is essential to support students in Cluster 2 and Cluster 3 throughout the semester and encourage higher activity levels. The first predicted timeframe will be Q1, representing the first three weeks of the course. The subsequent timeframe will be Q2, including Q1 plus the following three weeks, thus covering the first six weeks. Similarly, Q3 and Q4 will be proceeded with.

After employing the BIRCH algorithm to identify four distinct class labels, the clustered dataset has been utilized to develop a classification model tailored to predict student performance. This development focuses on selecting the optimal model to classify students based on accurately generated performance class labels.

To ensure the robustness and generalizability of the predictive models, 10-fold cross-validation was employed. This method systematically splits the dataset into 10 folds, with the model being trained on 9 folds and tested on the remaining fold. This process is repeated 10 times, ensuring that each part of the data contributes to both the training and testing phases. The performance of these classification models is systematically documented in Table 6. To enhance the applicability of the findings and enable proactive educational interventions, the prediction model was applied at several critical stages throughout the academic year. Specifically, predictions for the following timeframes were implemented: Q1, Q1-Q2, Q1-Q3, and Q1-Q4. This approach allows the assessment of student performance incrementally, facilitating timely interventions that can significantly alter academic outcomes. By analyzing performance at these successive intervals, educators can make informed decisions earlier in the academic process rather than awaiting end-of-course results. This strategy helps identify students who may require additional support early on. Simultaneously, it also aids in refining educational strategies and resource allocation to enhance overall student achievement.

Therefore, a comprehensive classification analysis was conducted to understand this relationship between student behavior, as recorded in log files, and their academic performance. The input dataset comprised six predictive features (*weekday, week, month, component, action, target)*extracted from the log files and target class representing a particular cluster.

SMOTE (Synthetic Minority Over-sampling Technique) was utilized to address significant class imbalances among the clusters. This was crucial as the initial data showed Cluster 0 having 6,647 instances and Cluster 1 with 6,456, whereas Cluster 2 had only 2,568 and Cluster 3 even fewer at 2,129. Using SMOTE enabled to enhance the representation of the minority classes, Cluster 2 and Cluster 3. The application of SMOTE not only balanced the class distribution but also enhanced the models' sensitivity to these underrepresented groups.

Subsequently, a range of machine learning models were applied, including Logistic Regression (LR), Random Forest (RF), Support Vector Machine (SVM), and K-Nearest Neighbors (KNN) to ascertain the likelihood of correctly predicting a student's cluster membership—a classification derived from their engagement patterns. Each model was meticulously tuned with a set of hyperparameters suited to the structure and complexity of the examined dataset. The main focus was on Clusters 2 and 3, which are crucial due to a higher incidence of potential dropouts among these students. Effective early intervention in these clusters can significantly impact student retention and success. As a result, the findings related to *Cluster 2* analysis model performance can be summarized as follows:

- The RF model exhibits a precision of 0.60 and a recall of 0.85, achieving an F1 score of 0.71. This suggests that while the model effectively identifies a high percentage of at-risk students (high recall), it also has a significant

**TABLE 5.** Results for performance attributes for each cluster.

| No. | Attribute | Cluster 0 | Cluster 1 | Cluster 2 | Cluster 3 |
|---|---|---|---|---|---|
| 1 | Submissions Q1 | *None*<br>*(0) 0.00%*<br>*1*<br>*(6) 16.14%*<br>*2*<br>*(18) 31.43%*<br>**3 or above**<br>**(11) 51.43%** | *None*<br>*(1) 2.94%*<br>*1*<br>*(9) 26.47%*<br>**2**<br>**(17) 50.00%**<br>*3 or above*<br>*(7) 20,59%* | *None*<br>*(8) 22.86%*<br>**1**<br>**(20) 57.14%**<br>*2*<br>*(7) 20.00%*<br>*3 or above*<br>*(0) 0.00%* | *None*<br>*(10) 19.23%*<br>**1**<br>**(35) 67.31%**<br>*2*<br>*(7) 13.46%*<br>*3 or above*<br>*(0) 0.00%* |
| 2 | Submissions Q2 | *None*<br>*(2) 5.71%*<br>*1*<br>*(3) 8.57%*<br>*2*<br>*(2) 5.71%*<br>**3 or above**<br>**(28) 80.00%** | *None*<br>*(4) 11.76%*<br>*1*<br>*(6) 17.65%*<br>*2*<br>*(3) 8.82%*<br>**3 or above**<br>**(21) 61.76%** | ***None***<br>***(24) 68.75%***<br>*1*<br>*(3) 8.57%*<br>*2*<br>*(6) 17.14%*<br>3 or above<br>(2) 5.71% | ***None***<br>***(35) 67.31%***<br>*1*<br>*(5) 9.62%*<br>*2*<br>*(11) 21.15%*<br>3 or above<br>(1) 1.92% |
| 3 | Submissions Q3 | *None*<br>*(1) 2.86%*<br>*1*<br>*(11) 31.43%*<br>*2*<br>*(10) 28.57%*<br>**3 or above**<br>**(13) 37.14%** | *None*<br>*(6) 17.65%*<br>*1*<br>*(8) 23.53%*<br>*2*<br>*(4) 11.76%*<br>**3 or above**<br>**(16) 47.06%** | *None*<br>*(5) 14.29%*<br>*1*<br>*(7) 20.00%*<br>***2***<br>***(12) 34.29%***<br>3 or above<br>(11) 31.43% | *None*<br>*(1) 1.92%*<br>*1*<br>*(17) 32.69%*<br>**2**<br>**(25) 48.08%**<br>*3 or above*<br>*(9) 17.31%* |
| 4 | Submissions Q4 | *None*<br>*(1) 2.86%*<br>*1*<br>*(2) 5.71%*<br>*2*<br>*(7) 20.00%*<br>**3 or above**<br>**(25) 71.43%** | *None*<br>*(2) 5.88%*<br>*1*<br>*(3) 8.82%*<br>*2*<br>*(3) 8.82%*<br>**3 or above**<br>**(26) 76.47%** | *None*<br>*(4) 13.33%*<br>*1*<br>*(2) 6.67%*<br>***2***<br>***(24) 80.00%***<br>3 or above<br>(0) 0.00% | ***None***<br>***(37) 71.15%***<br>*1*<br>*(13) 25.00%*<br>*2*<br>*(1) 1.92%*<br>*3 or above*<br>*(1) 1.92%* |
| 5 | Quizzes Q2 | *below 6.00*<br>*(0) 0.00%*<br>*6.00 – 7.00*<br>*(1) 2.86%*<br>*7.00 – 8.00*<br>*(7) 20.00%*<br>**8.00 – 10.00**<br>**(27) 77.14%** | *below 6.00*<br>*(0) 0.00%*<br>*6.00 – 7.00*<br>*(4) 11.76%*<br>**7.00 – 8.00**<br>**(22) 64.71%**<br>*8.00 – 10.00*<br>*(8) 23.53%* | *below 6.00*<br>*(10) 28.57%*<br>**6.00 – 7.00**<br>**(20) 57.14%**<br>*7.00 – 8.00*<br>*(5) 14.29%*<br>*8.00 – 10.00*<br>*(0) 0.00%* | **below 6.00**<br>**(23) 67.62%**<br>*6.00 – 7.00*<br>*(19) 26.54%*<br>*7.00 – 8.00*<br>*(6) 5.84%*<br>8.00 – 10.00<br>(0) 0.00% |
| 6 | Mid-term test Q3 | *below 9.00*<br>*(0) 0.00%*<br>*9.00 – 11.00*<br>*(12) 34,29%*<br>**11.00 – 13.00**<br>**(20) 57.14%**<br>*13.00 – 15.00*<br>*(3) 8.57%* | *below 9.00*<br>*(5) 14.71%*<br>*9.00 – 11.00*<br>*(11) 32.35%*<br>**11.00 – 13.00**<br>**(18) 52,94%**<br>*13.00 – 15.00*<br>*(0) 0.00%* | below 9.00<br>*(13) 37.14%*<br>**9.00 – 11.00**<br>**(20) 57.14%**<br>*11.00 – 13.00*<br>*(2) 5.71%*<br>*13.00 – 15.00*<br>*(0) 0.00%* | *below 9.00*<br>*(23) 44.23%*<br>**9.00 – 11.00**<br>**(27) 51,92%**<br>*11.00 – 13.00*<br>*(2) 3.85%*<br>*13.00 – 15.00*<br>*(0) 0.00%* |
| 8 | Grade | C<br>(17) 48.57% | D<br>(27) 79.41% | E<br>(35) 71,12% | Fx<br>(32) 61.54% |
| 9 | Dropout status | Pass<br>100% | Pass<br>100% | Pass<br>85.37% | Fail<br>61.54% |




**FIGURE 18.** Visualization of academic activity trends by quartile in identified clusters.

**TABLE 6.** Classification model scores for Q1 (3 weeks).

| Model | Hyperparameters | Class | Precision | Recall | F1 score | Accuracy | FP | FN |
|---|---|---|---|---|---|---|---|---|
| *RF* | *max_depth = 3*<br>*min_sample_leaf = 4*<br>*min_sample_split = 10*<br>*n_estimators = 100* | *Cluster 0* | *0.97* | *0.81* | *0.88* | **0.77** | 50 | 379 |
| | | *Cluster 1* | *0.79* | *0.76* | *0.77* | | 391 | 465 |
| | | *Cluster 2* | *0.60* | *0.85* | *0.71* | | 436 | 116 |
| | | *Cluster 3* | *0.80* | *0.66* | *0.72* | | 105 | 217 |
| *LR* | *C = 0.1*<br>*solver= liblinear*<br>*penalty = l2* | *Cluster 0* | *0.95* | *0.82* | *0.88* | **0.76** | 86 | 359 |
| | | *Cluster 1* | *0.78* | *0.75* | *0.76* | | 464 | 290 |
| | | *Cluster 2* | *0.61* | *0.85* | *0.71* | | 418 | 116 |
| | | *Cluster 3* | *0.80* | *0.65* | *0.72* | | 104 | 223 |
| *SVM* | *C = 1*<br>*kernel = linear*<br>*gamma = poly* | *Cluster 0* | *0.96* | *0.82* | *0.88* | **0.76** | 68 | 359 |
| | | *Cluster 1* | *0.78* | *0.75* | *0.77* | | 410 | 484 |
| | | *Cluster 2* | *0.61* | *0.83* | *0.70* | | 409 | 131 |
| | | *Cluster 3* | *0.79* | *0.66* | *0.72* | | 112 | 217 |
| *KNN* | *n_neighbors = 3*<br>*weights = distance*<br>*p = 3* | *Cluster 0* | *0.98* | *0.78* | *0.86* | **0.75** | 32 | 439 |
| | | *Cluster 1* | *0.77* | *0.77* | *0.77* | | 445 | 445 |
| | | *Cluster 2* | *0.57* | *0.90* | *0.70* | | 523 | 77 |
| | | *Cluster 3* | *0.85* | *0.59* | *0.70* | | 66 | 262 |

rate of false positives (436), indicating a potential for over-identification of students as at-risk.

- LR shows similar performance with a precision of 0.61, recall of 0.85, and an F1 score of 0.71. The model maintains a high sensitivity but with a considerable number of false positives (418), matching the trend observed in the case of RF.
- The SVM demonstrates a slightly better precision of 0.61 and a recall of 0.83, with an F1 score of 0.70. The model slightly reduces the false positive count to 409 but still identifies many students incorrectly as at-risk.
- KNN has the lowest precision among the models at 0.57, with a high recall of 0.90, leading to the highest false positives (523) but the lowest false negatives (77), indicating that while it is very good at detecting potential dropouts, it also misclassifies a considerable number of students.

In the case of Cluster 3, the analysis of model performance can be summarized as follows:

- The RF model achieves a precision of 0.80 and a recall of 0.66, resulting in an F1 score of 0.72. It has fewer false positives (105) compared to Cluster 2, suggesting a more conservative approach in flagging students as at-risk.
- LR shows similar figures with a precision of 0.80 and a recall of 0.65, achieving an F1 score of 0.72. This model also indicates a more cautious identification strategy with 104 false positives.
- SVM achieves a precision of 0.79 and a recall of 0.66, with an F1 score of 0.72. Like RF and LR, it shows a balanced approach with a moderate count of false positives (112).
- KNN presents the highest precision of 0.85 but the lowest recall of 0.59, reflected in an F1 score of 0.70. This model is the most conservative, with the fewest false positives (66), but misses many at-risk students, evidenced by the highest false negatives (262).

The early prediction results across all models indicate a crucial trade-off between identifying at-risk students and avoiding the misclassification of students who are not at risk. The high recall rates are promising for early intervention. However, the significant number of false positives, particularly in Cluster 2 across all models, suggests that while interventions are necessary, they need to be targeted and nuanced to avoid overburdening students who are not at risk. The application of SMOTE in this context helped to improve model learning on a limited number of data but also emphasizes the need for careful calibration of the intervention thresholds based on the predictive results.

For the initial six weeks of the semester (Q1-Q2), the classification model scores summarized in Table 7 reveal crucial insights into the predictive capabilities concerning students at higher risk of dropping out, particularly in Clusters 2 and 3. The data distribution within each cluster during this period was as follows: Cluster 0 had 14,428 students, Cluster 1 had 16,108, Cluster 2 had 9,942, and Cluster 3 had 7,106. From each cluster, 30% of the data was used for testing, resulting in test sizes of 4,328 for Cluster 0, 4,832 for Cluster 1, 2,983 for Cluster 2, and 2,132 for Cluster 3. Due to the

**TABLE 7.** Classification model scores for Q1-Q2 (6 weeks).

| Model | Hyperparameters | Class | Precision | Recall | F1 score | Accuracy | FP | FN |
|---|---|---|---|---|---|---|---|---|
| *RF* | *max_depth = 20*<br>*min_sample_leaf = 3*<br>*min_sample_split = 4*<br>*n_estimators = 150* | *Cluster 0* | *0.91* | *0.89* | *0.90* | **0.88** | 337 | *341* |
| | | *Cluster 1* | *0.80* | *0.86* | *0.83* | | 1039 | *612* |
| | | *Cluster 2* | *0.84* | *0.85* | *0.85* | | 471 | *423* |
| | | *Cluster 3* | *0.86* | *0.91* | *0.88* | | 259 | *192* |
| *LR* | *C = 0.01*<br>*solver= saga*<br>*penalty = l1* | *Cluster 0* | *0.96* | *0.83* | *0.89* | **0.87** | 149 | *727* |
| | | *Cluster 1* | *0.79* | *0.89* | *0.84* | | 1153 | *531* |
| | | *Cluster 2* | *0.79* | *0.92* | *0.85* | | 1267 | *239* |
| | | *Cluster 3* | *0.98* | *0.85* | *0.91* | | 37 | *320* |
| *SVM* | *C = 0.001*<br>*kernel = rbf*<br>*gamma = 0.01* | *Cluster 0* | *0.93* | *0.90* | *0.91* | **0.85** | 289 | *432* |
| | | *Cluster 1* | *0.80* | *0.82* | *0.81* | | 991 | *866* |
| | | *Cluster 2* | *0.90* | *0.75* | *0.81* | | 248 | *746* |
| | | *Cluster 3* | *0.81* | *0.92* | *0.86* | | 422 | *170* |
| *KNN* | *n_neighbors = 7*<br>*weights = uniform*<br>*p = 1* | *Cluster 0* | *0.93* | *0.90* | *0.91* | **0.84** | 289 | *432* |
| | | *Cluster 1* | *0.80* | *0.80* | *0.80* | | 966 | *966* |
| | | *Cluster 2* | *0.88* | *0.74* | *0.80* | | 251 | *813* |
| | | *Cluster 3* | *0.79* | *0.92* | *0.85* | | 485 | *170* |

underrepresentation of at-risk student population clusters, the SMOTE technique was also used. The findings, which will be summarized in the next paragraph, were shown by an in-depth analysis of the results for these two critical clusters.

*Cluster 2* Analysis of model performance:

- RF shows strong precision and recall (0.84 and 0.85, respectively) with an F1 score of 0.85. This balance suggests a reliable identification of at-risk students, though with a considerable number of FP = 471.
- LR exhibits lower precision (0.79) but high recall (0.92), resulting in an F1 score of 0.85. This model has a high sensitivity but is prone to a significant number of false positives (1267), suggesting it may overestimate the number of at-risk students.
- SVM with a precision of 0.90 but a lower recall of 0.75, the F1 score is 0.81. This model, while precise, fails to capture a significant portion of at-risk students, as indicated by 746 false negatives.
- KNN shows a high precision of 0.88 but a lower recall of 0.74. It identifies a high percentage of at-risk students with many false positives (FP = 491), which could impact resource allocation.

*Cluster 3* analysis of model performance:

- RF achieves higher precision and recall (0.86 and 0.91, respectively), leading to a superior F1 score of 0.88. The model effectively identifies at-risk students with fewer errors, evidenced by 259 false positives, which is lower compared to Cluster 2.
- LR shows excellent precision (0.98) with good recall (0.85), resulting in an F1 score of 0.91. This model is highly accurate in its predictions, with only 37 false positives, marking it as exceptionally efficient for Cluster 3.
- SVM with a precision (0.81) and recall (0.92) balance out to an F1 score of 0.86, making it a reliable model for identifying at-risk students, albeit with a moderate number of false positives (422).
- KNN achieved a precision that dips to 0.79 while maintaining a high recall of 0.92, culminating in an F1 score of 0.85. Despite the high sensitivity, the model incurs many false positives (485).

Across the models, varying degrees of effectiveness were observed in identifying at-risk students in Clusters 2 and 3. The RF model shows a strong balance between precision and recall, especially in Cluster 3. In contrast, the LR, particularly for Cluster 3, stands out for its precision, suggesting its application in environments where minimizing false positives is crucial. The SVM and KNN models highlight the challenges of balancing false positives and negatives, emphasizing the need for tailored strategies that consider the specific dynamics of each student cluster.

The evaluation of classification models (Table 8 ) over the initial nine weeks of the semester encompassing the first three quarters (Q1 through Q3). These periods mark significant phases in the academic timeline, each adding a layer of data richness that enhances the predictability and reliability of the models.

Specifically, for this analysis, the following amount of data points was available: 27,787 for cluster 0, 32,416 for cluster 1, 24,512 for cluster 2, and 36,406 for cluster 3. Again, 30% of the data were selected for the test set from each class, resulting in sample sizes 8,336 for Cluster 0, 9,725 for

**TABLE 8.** Classification model scores for Q1-Q3 (9 weeks).

| Model | Hyperparameters | Class | Precision | Recall | F1 score | Accuracy | FP | FN |
|---|---|---|---|---|---|---|---|---|
| *LR* | *C = 0.1*<br>*solver= sag*<br>*penalty = l1* | *Cluster 0* | *0.97* | *0.94* | *0.96* | **0.95** | 239 | 500 |
| | | *Cluster 1* | *0.92* | *0.88* | *0.90* | | 699 | 1167 |
| | | *Cluster 2* | *0.91* | *0.97* | *0.94* | | 678 | 220 |
| | | *Cluster 3* | *0.95* | *0.97* | *0.96* | | 559 | 324 |
| *RF* | *max_depth = 25*<br>*min_sample_leaf = 3*<br>*min_sample_split = 5*<br>*n_estimators = 120* | *Cluster 0* | *0.97* | *0.98* | *0.97* | **0.94** | 251 | 167 |
| | | *Cluster 1* | *0.90* | *0.93* | *0.91* | | 1005 | 681 |
| | | *Cluster 2* | *0.96* | *0.91* | *0.94* | | 278 | 662 |
| | | *Cluster 3* | *0.93* | *0.94* | *0.93* | | 793 | 656 |
| *SVM* | *C = 1*<br>*kernel = sigmoid*<br>*gamma = auto* | *Cluster 0* | *0.95* | *0.94* | *0.95* | **0.92** | 412 | 500 |
| | | *Cluster 1* | *0.81* | *0.90* | *0.85* | | 2058 | 972 |
| | | *Cluster 2* | *0.93* | *0.84* | *0.88* | | 462 | 1167 |
| | | *Cluster 3* | *0.98* | *0.95* | *0.97* | | 211 | 546 |
| *KNN* | *n_neighbors = 10*<br>*weights = uniform*<br>*p = 3* | *Cluster 0* | *0.97* | *0.98* | *0.97* | **0.91** | 251 | 167 |
| | | *Cluster 1* | *0.90* | *0.93* | *0.91* | | 1005 | 681 |
| | | *Cluster 2* | *0.96* | *0.91* | *0.94* | | 278 | 662 |
| | | *Cluster 3* | *0.93* | *0.94* | *0.93* | | 793 | 656 |

Cluster 1, 7,354 for Cluster 2, and 10,922 for Cluster 3. Given the relatively equal distribution among individual classes, the sampling approach was deemed unnecessary.

*Cluster 2* analysis of model performance:

- LR demonstrated high precision (0.91) and excellent recall (0.97), leading to an F1 score of 0.94 and an accuracy of 0.95. However, it had a significant number of false positives (678).
- RF achieved high precision (0.96) and very good recall (0.91), resulting in an F1 score of 0.94 and matching accuracy. False positives were lower (278) than LR, but false negatives were considerably higher (662).
- SMV showed high precision (0.93) but lower recall (0.84), leading to an F1 score of 0.88 and an accuracy of 0.92. It had a moderate number of false positives (462) but a significant number of false negatives (1167).
- KNN exhibited exceptional precision (0.96) and good recall (0.91), with an F1 score of 0.94 and accuracy of 0.91. This model had relatively few false positives (278) but many false negatives (662).

*Cluster 3* analysis of model performance:

- LR showed even higher precision (0.95) and recall (0.97), achieving an F1 score of 0.96. This model had fewer false positives (559) than Cluster 2 but more false negatives (324), indicating a slightly less balanced but still effective prediction capability.
- RF resulted in slightly lower precision (0.93) and similar recall (0.94) with an F1 score of 0.93 and the same accuracy. Like Cluster 2, it had fewer false positives (793) but many false negatives (656).
- SMV excelled with high precision (0.98) and high recall (0.95), achieving an F1 score of 0.97. It had far fewer false positives (211) and false negatives (546), indicating a highly effective model for this cluster.
- KNN achieved lower precision (0.93) and excellent recall (0.93), resulting in an F1 score of 0.93. The number of false positives was higher (793), matching the false negatives (656).

Across all models, Clusters 2 and 3 exhibit variations in performance, with certain models excelling in precision while others in recall. The LR model was notably effective in both clusters for maximizing recall, which is crucial for the early identification of at-risk students. However, the high number of false positives suggests that while it captures most at-risk students, it also incorrectly flags some who may not require interventions. The RF and KNN models demonstrated a good balance between precision and recall. Still, they suffered from a high number of false negatives, potentially missing a substantial number of at-risk students. Conversely, the SVM model, particularly in Cluster 3, stood out for its high precision and recall, suggesting its suitability for applications where both accurate identification and comprehensive coverage of at-risk students are critical. These results underscore the importance of choosing a suitable model based on the specific needs of the educational setting and the characteristics of the student behavior being analyzed.

Finally, the dataset was also thoroughly analyzed to assess student performance across the four clusters, encompassing all quarters up to Q4. The distribution for this entire semester period was as follows: Cluster 0 had 37,996 data points,

**TABLE 9.** Classification model scores for Q1 – Q4 (13 week).

| Model | Hyperparameters | Class | Precision | Recall | F1 score | Accuracy | AUC ROC | FP | FN |
|---|---|---|---|---|---|---|---|---|---|
| *LR* | *C = 1*<br>*solver= lbfgs*<br>*penalty = l2* | *Cluster 0* | *0.98* | *0.95* | *0.96* | **0.96** | 0.98 | 325 | 136 |
| | | *Cluster 1* | *0.95* | *0.90* | *0.92* | | 0.96 | 316 | 699 |
| | | *Cluster 2* | *0.94* | *0.99* | *0.96* | | 0.97 | 393 | 210 |
| | | *Cluster 3* | *0.99* | *0.98* | *0.98* | | 0.98 | 30 | 19 |
| *RF* | *max_depth = 6*<br>*min_sample_leaf = 2*<br>*min_sample_split = 2*<br>*n_estimators = 100* | *Cluster 0* | *0.98* | *0.95* | *0.96* | **0.95** | 0.98 | 282 | 553 |
| | | *Cluster 1* | *0.95* | *0.90* | *0.92* | | 0.94 | 559 | 1312 |
| | | *Cluster 2* | *0.92* | *0.98* | *0.96* | | 0.95 | 1044 | 96 |
| | | *Cluster 3* | *0.98* | *0.96* | *0.97* | | 0.99 | 33 | 16 |
| *SVM* | *C = 0.1*<br>*kernel = rbf*<br>*gamma = poly* | *Cluster 0* | *0.97* | *0.94* | *0.95* | **0.95** | 0.96 | 396 | 617 |
| | | *Cluster 1* | *0.90* | *0.92* | *0.91* | | 0.95 | 958 | 1189 |
| | | *Cluster 2* | *0.94* | *0.95* | *0.94* | | 0.97 | 812 | 376 |
| | | *Cluster 3* | *0.99* | *0.98* | *0.98* | | 0.98 | 35 | 19 |
| *DT* | *max_depth = 5*<br>*min_sample_leaf = 2*<br>*min_sample_split = 2*<br>*max_features = sqrt* | *Cluster 0* | *0.98* | *0.96* | *0.97* | **0.94** | 0.97 | 284 | 496 |
| | | *Cluster 1* | *0.95* | *0.91* | *0.93* | | 0.96 | 553 | 1066 |
| | | *Cluster 2* | *0.96* | *0.95* | *0.94* | | 0.98 | 800 | 33 |
| | | *Cluster 3* | *0.94* | *0.93* | *0.93* | | 0.95 | 33 | 16 |
| *KNN* | *n_neighbors = 5*<br>*weights = distance*<br>*algorithm = auto*<br>*p = 2* | *Cluster 0* | *0.95* | *0.94* | *0.95* | **0.94** | 0.96 | 569 | 643 |
| | | *Cluster 1* | *0.91* | *0.90* | *0.90* | | 0.93 | 1243 | 1144 |
| | | *Cluster 2* | *0.92* | *0.94* | *0.93* | | 0.97 | 1001 | 714 |
| | | *Cluster 3* | *0.98* | *0.96* | *0.97* | | 0.98 | 113 | 425 |

Cluster 1 had 39,455, Cluster 2 had 35,785, and Cluster 3 had 37,645. Upon evaluating the last period, the models confirmed promising results, as summarized in Table 9. As was expected, the classification report for each model reveals robust precision, recall, and F1 scores across all clusters, indicating a high level of consistency in the models' predictive accuracy. Notably, the AUC ROC values, which serve as a benchmark for classification performance where a score of 1 represents a perfect model, were impressive across the board, reflecting each model's strong discriminative ability.

Upon analyzing the rates of false positives and false negatives for each cluster, variability was observed, which provides insightful nuances in the prediction capabilities of each model. These findings underscore the viability of employing machine learning algorithms to forecast cluster membership based on student log data. At the same time, this predictive modeling approach confirms the relationship between engagement patterns and academic outcomes. Moreover, it emphasizes the possibility of using such analyses to customize educational interventions and support mechanisms that are responsive to students' various behaviors during their learning process.

Determining the 'best' classifier involves a nuanced consideration of these metrics in the context of the specific application and priorities. Logistic Regression and Random Forest demonstrated a higher recall, effectively identifying a broader set of at-risk students. This characteristic is crucial for interventions aimed at minimizing the number of students who might drop out unnoticed. However, the high recall comes with lower precision in the Q1 and Q2 periods, which is associated with higher FP, meaning that these models may also incorrectly classify some students as at-risk. This could lead to over-intervention, where resources might be allocated to students who do not actually require help.

In contrast, SVM and KNN focused on high precision, which results in fewer FP but higher FN due to low recall values. This trade-off suggests that while they effectively target students who are at risk, they may exclude some students who could benefit from interventions.

In conclusion, understanding the strengths and limitations of each model in terms of recall and precision allows educational institutions to tailor their predictive analytics strategies effectively. This tailoring ensures that interventions are appropriately targeted and aligned with the overarching goals of supporting student success in the most resource-efficient manner possible.

## V. CONCLUSION

The research's findings emphasize the noticeable impact of student engagement on academic performance. The BIRCH algorithm's ability to classify students into clusters based on their LMS Moodle activity levels underscores the predictive power of engagement metrics. A nuanced understanding emerges from the distinct clusters:

- **Cluster 0** represents consistent and thorough student engagement associated with higher grades.
- **Cluster 1** indicates moderate engagement and average performance.
- **Cluster 2** is characterized by activity centered around mid-term assessments and is linked to borderline passing grades.
- **Cluster 3** shows minimal engagement and is associated with failing grades and a high dropout rate.

The discussion of findings revealed by clustering algorithms elucidates the diverse student engagement strategies within a blended learning course. Notably, BIRCH's effective segmentation of student behaviors correlates closely with academic achievements, from consistent engagement to minimal interaction patterns. While BIRCH exemplifies algorithmic precision in capturing student engagement, the nuanced performance of DBSCAN and GMM highlights the complexity of student behavior analytics.

Limitations of this study include its concentration on a single academic institution and course, which may restrict the generalizability of the findings. When analyzing multiple courses with different structures across various disciplines, it becomes challenging to identify consistent or useful patterns of student behavior, as each course may exhibit distinct dynamics. Therefore, as taken in this study, a more focused approach allows deeper insights into individual course dynamics and offers to refine course structures to better capture student behavior. However, one approach that could enhance the predictive model without incorporating personal data would be considering a student's performance in other courses as a contextual factor. This information could be leveraged to refine predictions in the course under study, providing a more comprehensive view of a student's academic trajectory based solely on their online traces. In addition, using log files assumes that the records accurately and comprehensively capture a student's activity.

Despite these limitations, the study provides a framework that uses LMS data to predict academic achievement. This framework gives educators a proactive tool to identify students who may be in danger of underperforming or dropping out.

Furthermore, as educational datasets continue to grow, particularly with the increasing shift towards online and hybrid learning environments, the scalability of the chosen clustering algorithms becomes a critical consideration. BIRCH, by design, scales well with large datasets, making it particularly suited for growing data environments. On the other hand, DBSCAN's computational complexity may hinder its performance when faced with extremely large datasets, as calculating distances between data points can become intensive. GMM, with its probabilistic nature, may also encounter computational constraints as dataset size increases. Future research should explore how these algorithms perform in larger-scale educational settings, and the development of more scalable clustering methods could enhance their practical application for institutional-level data analytics.

The generalizability of the findings is another important factor. Since the study was conducted within the context of a single institution and course, future studies should consider replicating this methodology across different institutions and learning environments to better understand how student engagement patterns differ. However, the use of a standard LMS Moodle log file in this experiment not only offers a solid baseline for future studies but also ensures reproducibility thanks to its widely accepted format. This will help establish whether the insights from clustering algorithms like BIRCH, DBSCAN, and GMM hold true across various educational contexts, improving the applicability of the findings to diverse educational settings.

Future research could benefit from a broader dataset that includes data collected over multiple teaching periods of the same course, resulting in larger cohorts. This would help contextualize their engagement in the specific course being studied and offer a more robust predictive model without compromising the focus on online traces. Longitudinal studies would also be beneficial in tracking changes in student engagement over multiple teaching periods, providing further insights into the stability and adaptability of the predictive model. Additionally, as the nature of learning environments evolves, particularly with the rise of hybrid and fully online formats, it will be valuable to assess whether the same engagement patterns identified in this study apply to those contexts. This would ensure that predictive models are adaptable to various learning formats and can be used to address emerging challenges in student engagement and retention.

Nonetheless, this research's contribution is substantial, offering a predictive framework that identifies at-risk students using clustering and classification and facilitates timely educational interventions. This study's findings underscore the importance of a systematic approach to monitoring student engagement through learning management systems. The BIRCH algorithm's effective clustering allows for a predictive model that aligns closely with academic performance. The confirmation of consistent engagement correlating with higher academic achievement and sporadic interaction with lower outcomes calls for educational strategies that promote regular LMS usage. The drawn conclusion emphasizes the potential of machine learning in shaping educational policies and supporting students' academic journeys, reflecting on the critical role of data-driven interventions in enhancing the quality of higher education.

Conclusively, this study advances the comprehension of the link between student LMS engagement and academic outcomes. By applying clustering algorithms, it was demonstrated that predicting a student's likelihood of academic success or failure is possible based on their digital footprints within an LMS. The BIRCH algorithm, in particular, showed superior performance in delineating student groups with different academic trajectories. By identifying engagement patterns that indicate academic success, educators and administrators can tailor support to students in a more targeted and effective manner. Moreover, the potential of these

predictive models to be integrated into early warning systems within LMS platforms can help institutions take a proactive approach in reducing dropout rates and improving student performance. Furthermore, the predictive model derived from this research offers a tool to pre-emptively address challenges in student performance, thereby contributing to the broader goals of educational success and student dropout.

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